

Bucco — A Scientific Field Guide to His First Year

Compiled for Scott Smathers • North Hills, Pittsburgh

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Bucco

A Scientific Field Guide to His First Year

Border Collie puppyhood through the lens of behavioral neuroscience, learning theory,
nutrition, and breed genetics

Compiled for Scott Smathers

North Hills, Pittsburgh • May 2026

About This Book

This is a working scientific reference on Bucco's first year — built for Scott to read straight through, jump around in, or hand to Nina. The orientation: take what's known from canine ethology, neuroscience, learning theory, and Border Collie-specific behavioral research, and translate it into specific weekly actions and watchouts for one puppy in one Pittsburgh apartment.

The seven chapters that follow:

1. **Week-by-Week.** The cornerstone. A developmental atlas from week 12 (now) through week 52, with expected weight, calorie targets, training goals, socialization checkpoints, and watchouts for every week.
2. **The Neuroscience of Puppyhood.** What is actually happening in his brain at each stage — critical periods, fear windows, sleep architecture, myelination, classical and operant conditioning at the neural level.
3. **Feeding Science.** The real numbers behind the bowl. RER calculations, macronutrients, calcium/phosphorus ratios, the grain-free/DCM situation, the microbiome angle, and the food-as-training-currency philosophy.
4. **Learning Theory.** Susan Garrett's method, anchored in the behavioral science that justifies each piece. Why high reinforcement rates work, why "no" doesn't, why classical conditioning has to precede operant.
5. **Border Collie Genetics & Behavior.** What makes a BC a BC. The Coppinger predatory motor pattern, the four major genetic diseases, compulsive disorder risk, BC adolescence specifically, and Chaser the 1,022-word dog.
6. **Fun Facts & Research.** A grab-bag of canine cognition, sensory neuroscience, behavioral genetics, and BC-specific trivia.
7. **Warnings & Red Flags.** What never to do, what to watch for, and when to call which kind of professional.

The whole book runs through one filter: science, applied specifically to Bucco, in the spirit of the Susan Garrett framework. Nothing here contradicts the existing first-year training plan; everything here gives that plan its biological backbone.

A note on sources. Every chapter ends with a sources section. Many of the underlying claims trace back to peer-reviewed work — Scott & Fuller (1965), Coppinger & Coppinger (2001), the Pilley/Reid work on Chaser, the Iltis sleep electrophysiology papers, the Hart et al. UC Davis spay/neuter studies, Hazewinkel's calcium work, Pavlov, Skinner, Premack, Pryor. Practical sources (training organizations, veterinary medical sites, breed clubs) are also cited where they synthesize current consensus reliably.

Chapter 1 • The Week-by-Week Atlas

A developmental atlas from week 12 (where we are now) through week 52. For each week: what's happening in his brain and body, expected weight band, calories, training goals, socialization checkpoints, vet milestones, and what to watch for.

Reference dates (Bucco DOB: February 18, 2026)

- Week 12 starts: May 11, 2026 (today's reference)
- Week 16: June 8, 2026 (final puppy boosters scheduled June 11)
- Week 20: July 6, 2026
- Week 26 (6 months): August 17, 2026
- Week 39 (9 months): November 16, 2026
- Week 52 (12 months): February 15, 2027

How to read the weight bands. Border Collies vary widely. The chart below uses literature-derived growth curves for medium-line BCs (35-50 lb adult). Working-line dogs tend to run lean and finish at the lower end. Use the band as a *check*, not a target — body condition score (BCS) is the real metric.

How to read the calorie targets. Calculated from the RER formula:

$$\text{RER} = 70 \times (\text{kg}^{0.75}); \text{ puppy multiplier } \times 3 \text{ under 4 months, } \times 2 \text{ from 4-12 months, } \times 1.6 \text{ over 12 months}$$

These are *starting points*. Adjust by body condition every 2 weeks.

PHASE 1 — Primary Socialization Wraps Up (Weeks 12-16)

Week 12 (May 11-17, 2026) — WHERE WE ARE NOW

Body

- Expected weight: 12-18 lb (5.4-8.2 kg)
- Daily calories: ~745-1015 kcal (4 meals)
- Heart rate elevated, panting still inefficient — heat sensitivity high

Brain

- *Tail end of primary socialization window.* The visual cortex is finalizing its template for “what is a social partner.” What he meets and likes now sticks; what he doesn’t meet may become “scary forever.”
- REM sleep ~30% of total sleep (vs. ~10-12% in adult dogs). Memory consolidation is happening overnight.
- Myelination of cortical association areas still in progress. Impulse control is *physiologically* not available — the wiring isn’t there yet.

Training goals (the 30-day three)

1. **Crate Games** — 5 min/day. Door opens = stay. Release word = freedom. Build duration in 1-2 sec increments.
2. **Reinforcement Zone (RZ)** — Pick a side. Feed *into the zone*, never reach to him.

3. **Restrained recall** — Nina holds, Scott runs and calls, Nina releases, party at the finish line.

Plus: name recognition, IYC (It's Yer Choice) closed-fist, cooperative care pairing.

Socialization checkpoints (aim for 5+ new positive exposures this week)

- New surface (gravel? metal grate? wobble cushion?)
- New sound at meal time (vacuum at distance, then closer)
- New person (different age/silhouette than yesterday)
- Car ride that *doesn't* end at the vet
- Five minutes of crate-decompression in a new environment (carry in, blanket down, feed kibble)

Vet / health

- 12-week shots [done] done May 11
- Lyme [done] done May 11
- Heartgard chewable [done] started May 11
- **Set reminder: Schedule June 11 booster appointment**

Watchouts

- *Tail end of First Fear Period (8-11 weeks)*. Anything scary now imprints harder than usual. Soft hands, more space, more food. If he startles, do NOT comfort and coo — that reinforces the worry. Stay calm, create distance, feed when he reorients.
- Don't leash-correct. Don't say "no." Both undermine the trust you're building.
- Cats: Bucco MUST be leashed when Kiki and Rainier are loose. Every successful stalk rehearsal makes the next one likelier.

Why this week matters

The puppy you meet in adolescence is built from these weeks. Errors compound; wins compound harder.

Week 13 (May 18-24, 2026)

Body

- Expected weight: 14-21 lb (6.4-9.5 kg). Should be gaining ~2 lb/week now.
- Calories: ~870-1085 kcal (still 4 meals)
- Teething starting in earnest — adult teeth begin to push baby teeth. Expect chewing, drool, mild fussiness.

Brain

- Socialization window narrows. Anything novel must be *positive or neutral*; bad exposures stick.
- Sleep needs still 18-20 hours.

Training goals

- Crate Games duration to ~10 sec with door open
- Add Hand Target (nose-to-palm) — high reinforcement rate, 20+/min

- Begin food bowl gameplay for resource-guarding *prevention* (walk by, drop better food in, walk away)

Socialization

- First exposure to busy parking lot (carried, treats while he watches from a distance)
- Pair the doorbell with cheese — 5 reps a day
- Sound desensitization: thunder, fireworks, sirens at LOW volume during meals

Vet

- Confirm June 11 appointment is scheduled

Watchouts

- Teething nips. Provide frozen washcloth, cold Kong, raw carrot. Redirect mouthing to a tug toy. Never say “ouch!” loud — many BCs interpret as squeal of prey.
-

Week 14 (May 25-31, 2026)

Body

- Expected weight: 15-23 lb (6.8-10.5 kg)
- Calories: ~935-1170 kcal
- Coordination noticeably improving — running, jumping, but joints still very vulnerable.

Brain

- *Socialization window officially closing.* By end of this week he’s locking in his worldview.
- Brain weight ~70-80% of adult. The structure exists; the wiring continues.

Training goals

- Crate Games — door opening/closing without release. Duration to 20+ seconds.
- Begin Sit / Down position discrimination — *no luring after rep 3-5*. Verbal cues only.
- First “settle on mat” intro — 2 seconds, food rains down on the mat.

Socialization

- Vet visit dry run (drive to vet, sit in parking lot, eat treats, leave). Build positive emotional association *before* the June 11 needle visit.
- Find a sound or surface from your 100-thing checklist he hasn’t met.

Watchouts

- **No stairs, no jumping off furniture.** Growth plates are wide open. A bad landing this month = orthopedic problem at year 3.
 - No forced exercise. Free play yes; leash hikes no.
-

Week 15 (June 1-7, 2026)

Body

- Expected weight: 17-25 lb (7.7-11.4 kg)
- Calories: ~1020-1235 kcal
- Adult premolars/molars emerging. Heavy chew need.

Brain

- Last week of primary socialization (give or take). Cram in everything you can — within reason, never overwhelm.

Training goals

- Reinforcement Zone in motion: 2-3 steps, feed in zone. Foundation of loose-leash walking.
- Hand target across the room — moving target.
- Default sit at thresholds (doors, crate openings) — capture and reinforce, never command yet.

Socialization

- Visit a friend's clean, fully vaccinated adult dog — short, structured interaction
- New environment: outdoor café patio at off-peak hours, on a mat at your feet

Vet

- Confirm 16-week appointment (June 11) — what's included? Likely DHPP final + rabies. Ask about leptospirosis and bordetella based on Pittsburgh exposure.

Watchouts

- Resource guarding prevention is now: trade up. If he has something, swap it for something better, then give the original back. Never just take.

Week 16 (June 8-14, 2026) — [!] KEY MILESTONE

Body

- Expected weight: 18-27 lb (8.2-12.3 kg)
- Calories: ~1085-1300 kcal
- **June 11: 16-week vet visit — final puppy boosters + likely rabies.**

Brain

- Primary socialization window closes around now (some research extends it to 14-16 weeks, but 16 is the conservative end). After this, *new* fears form much more easily than they did before.

Training goals

- Crate Games: door fully open for 10+ seconds, no release until cue
- IYC progression: open palm with food on the ground, hand stays nearby. He must back off.
- Recall: still restrained recall only. Don't test "Bucco come!" without restraint yet.

Socialization

- The June 11 vet visit IS your training session. Treat-feed through the door, on the scale, on the table, during the exam. The vet should be the second-most-reinforcing place in his life.
- After: 2-week countdown to full immunity begins.

Vet

- 16-week shots [done] (after appointment)
- Monthly Heartgard [done] (next dose due ~June 11)
- Discuss spay/neuter timing — push for 12-18 months minimum (see §spay-neuter)

Watchouts

- Vaccine reactions: lethargy, mild swelling at injection site, low-grade fever for 24 hr is normal. *Hives, facial swelling, vomiting, collapse = ER now.*
 - The 24-48 hr post-vaccine: keep it quiet. Cuddle, sleep, calm food. No training sessions; no big social outings.
-

PHASE 2 — Engagement & Generalization (Weeks 17-20)

Week 17 (June 15-21, 2026)

Body

- Expected weight: 19-28 lb (8.6-12.7 kg)
- Calories: ~1100-1340 kcal
- *Transitioning to puppy multiplier* ×2 by week 18 — you'll start to see the calorie target stop climbing.

Brain

- Post-vaccine recovery week. Likely sleepier than usual the first 48 hr.
- Working memory expanding — he can now hold a 2-step task ("sit, then watch me").

Training goals

- Begin Tug as Training Currency. Two rules: only when offered, releases on cue.
- Hand target with movement: 3-4 steps following a target.
- Capture "look at me" anywhere he offers eye contact unprompted. This becomes the default attention behavior.

Socialization

- Still 1 week from full immunity. Continue with known dogs only.
- Sound work: turn the volume up 1-2 notches on the desensitization playlist.

Watchouts

- Some puppies show transient regression post-vaccine for 3-5 days (housetraining accidents, less engagement). Normal. Maintain routine, don't pile on training.
-

Week 18 (June 22-28, 2026)

Body

- Expected weight: 20-30 lb (9.1-13.6 kg)
- Calories: shift to ~750-985 kcal (×2 multiplier kicks in around 16-18 weeks; growth slowing)

- Recheck body condition — he should be feel-able ribs, visible waist from above.

Brain

- Frontal cortex development accelerates. Impulse control wiring starts to become *physiologically possible* (won't be *reliable* until 12-18 months).

Training goals

- “Drop” / “Out” of tug — never stop the game on the drop; restart immediately as reward.
- IYC with toys: keep the toy in your hand, he must offer disengagement before he gets to play.
- 4 meals → 3 meals by end of this week.

Socialization

- [!] **Full immunity** (~2 weeks post-final shots, so ~June 25). You can now access pet store floors, structured puppy classes, broader environments.
- Find a positive-reinforcement-only puppy class in Pittsburgh. Susan Garrett’s DogsThat referral network or KPA-CTP / CPDT-KA certified trainers.
- Still NO dog parks. Ever. (See \$warnings.)

Watchouts

- This is when many owners “graduate” off the careful socialization plan because the dog is “fine.” It’s not graduation; it’s the start of the harder work. Generalization (the same skill in new places) needs deliberate training.

Week 19 (June 29 - July 5, 2026)

Body

- Expected weight: 22-32 lb (10-14.5 kg)
- Calories: ~785-1020 kcal (3 meals)

Brain

- Mid-puppy plateau in some BCs. Energy may seem to dip for 3-5 days; this is real, not concerning.

Training goals

- Three-position discrimination (sit/down/stand) at home, no luring.
- Find It / scatter feeds — 5 min/day. Builds nose engagement and burns brain.
- Settle on mat to 30 sec.

Socialization

- First “field trip” to a quiet outdoor area he can walk on his own paws (not just be carried).
- Long-line work begins. 15-30 ft biothane. Recall on the long line in low-distraction outdoor space.

Watchouts

- Loose stool from new food, new water, new environment is common. Persistent diarrhea >48 hr = vet, especially under 6 months.

Week 20 (July 6-12, 2026)

Body

- Expected weight: 23-34 lb (10.5-15.4 kg)
- Calories: ~820-1055 kcal

Brain

- Coordination noticeably better. BCs at this age often start showing the eye-stalk-chase predatory pattern more visibly. *This is the signal to start managing it deliberately* (see §bc-genetics).

Training goals

- Loose-leash walking in driveway/quiet sidewalk — 30 sec stretches in RZ.
- Begin Two-Toy game (predecessor to formal retrieve).
- Distraction Intensity Index (DII): rate his environments 1-10. Train known behaviors at DII 1-3.

Socialization

- Group class, if you've found one. Otherwise: structured 1:1 with a known calm adult dog.
- New environment per week target: at least one.

Vet

- Heartgard monthly [done] (around July 11)

Watchouts

- BCs may show first signs of light/shadow fixation — staring at reflections, headlights on the ceiling, fan shadows. **Interrupt immediately and redirect to a high-engagement game.** Never play with reflections, never use a laser pointer.

PHASE 3 — Distraction Layering & Building Skills (Weeks 21-26)

Week 21 (July 13-19)

Body

- 25-36 lb (11.3-16.3 kg); ~860-1090 kcal
- Adult coat blowing in. Brush daily.

Brain

- BCs entering early juvenile period. Cognitive load capacity is rising fast.

Training goals

- Begin generalizing “sit” / “down” to backyard, then sidewalk. Same cue, new location.
- Recall on long line in DII-3 environments.
- Add “leave it” using IYC framework — never use “no.”

Week 22 (July 20-26)

Body

- 26-38 lb (11.8-17.2 kg); ~895-1125 kcal

Brain

- Object permanence solid. Hide-and-seek games work as enrichment now.

Training goals

- Settle on mat to 1 minute with mild distractions.
 - Tug with “drop” and “take it” both on cue.
 - Begin chin rest as cooperative-care skill (medical exam foundation).
-

Week 23 (July 27 - Aug 2)

Body

- 27-39 lb (12.2-17.7 kg); ~925-1155 kcal

Brain

- Sleep needs dropping toward 16-18 hr/day. He may seem more “on” — this is real, not behavioral.

Training goals

- Loose-leash walking in residential block. Pivot or about-face when he forges.
 - Distraction layering: ask known cues with a dog visible at 50 ft. Reinforce the response, not the behavior of “ignoring the dog” — that comes from how you feel about dogs, which is built by classical conditioning.
-

Week 24 (Aug 3-9)

Body

- 28-40 lb (12.7-18.1 kg); ~955-1185 kcal

Brain

- Approaching adolescence. Some BCs start the early hormonal shifts as soon as 5 months.

Training goals

- Capture default offered behaviors (sit when bored, down when relaxed). Reinforce silently — no marker, no fuss; just food appears.
-

Week 25 (Aug 10-16)

Body

- 29-42 lb (13.2-19 kg); ~985-1215 kcal

Brain

- Recheck the socialization log. Have you hit 100 positive exposures yet? If not, push the easy ones this week.

Training goals

- First recall test in DII-5 environment (quiet park, no other dogs). Long line on, but slack.
-

Week 26 (Aug 17-23) — [!] 6 MONTHS

Body

- 30-43 lb (13.6-19.5 kg); ~1015-1240 kcal
- 6-month vet checkup. Adult teeth fully in by now (final molars erupt around 6-7 months).

Brain

- *Adolescence officially begins around now. Brace yourself.*
- The Asher et al. (2020) Royal Society Biology Letters study: “Trainability” scores from owners drop *only* with their own dog at adolescence — not with strangers or new handlers. He **WILL** “forget” what he knows. This is socially specific, not cognitive.
- Prefrontal cortex (impulse control, decision-making) is still under construction. Amygdala (emotion, reactivity) is fully online. This imbalance is why teenagers (human and canine) are like this.

Training goals

- Maintain everything from Phase 1-3 at low rates of reinforcement (~5/min in known contexts).
- Hold the line on Crate Games, IYC, RZ. Don’t add new things this month — protect what works.

Vet

- 6-month wellness check. **Discuss spay/neuter — strongly advocate for waiting until 12-18 months for growth plate closure and reduced orthopedic/cancer risk.** See §spay-neuter.
- Heartgard monthly [done]

Watchouts

- [!] First wave of adolescent regression. He’ll suddenly act like he’s never heard “sit” before. Don’t escalate. Don’t get louder. Don’t punish. Ride it out with management and high reinforcement.
 - BC-specific watch: stress vocalization (whining, barking at nothing). Could be hormonal; could be early reactivity. Note frequency, context.
-

PHASE 4 — Adolescence Begins (Weeks 27-39)

Week 27 (Aug 24-30)

Body

- 31-44 lb (14-20 kg); ~1040-1265 kcal

Brain

- Estrogen/testosterone rise sharply. Emotional reactivity climbs.

Training goals

- Keep training sessions SHORT (2-3 min). Adolescent BCs frustrate fast.
 - Premack: use environmental rewards. "Sit before the door opens." "Look at me before we get to sniff that bush."
-

Week 28-31 (Aug 31 - Sep 27) — The Hormonal Storm

Body

- 31-46 lb plateau through this stretch; ~1040-1290 kcal
- 7 months hits in week 30 (mid-September).

Brain

- Trainability dip in full swing. He's not being stubborn — his brain is genuinely different right now.

Training goals

- Maintain. Maintain. Maintain.
- One concept per session.
- Stop sessions while he still wants more (this was always the rule, but here it's law).

Watchouts

- **Catch tiny early warning signs of:** resource guarding (lip lift, freeze over food), reactivity (lunging on leash), compulsion (light-staring, shadow-staring, flank-licking). All four are *vastly* easier to address in the first 2 weeks than the first 6 months.
-

Week 32 (Sep 28 - Oct 4)

Body

- 32-46 lb (14.5-20.9 kg); ~1055-1305 kcal
- ~7.5 months. Some males show subtle territorial marking.

Brain

- *Second fear period possible* (6-14 months, but typically 8-11 months for medium breeds). Watch for sudden new fears of things he was fine with last week.

Training goals

- Capture confidence. Reinforce approach behaviors, calm exploration.
-

Week 33-36 (Oct 5 - Nov 1) — 8 Months

Body

- ~33-47 lb through this stretch; ~1075-1330 kcal

Brain

- This is the *peer-reviewed adolescence trough* (Asher et al.). If trainability is going to dip, it dips here.

Training goals

- Don't introduce new skills. Maintain the ones he knows in *easier* environments than he could handle a month ago.
- Heavy classical conditioning work: pair scary/exciting things with food at a distance. How he *feels* matters more than what he does.

Watchouts

- **[!] Second Fear Period most likely now.** A confident puppy suddenly stops dead at a manhole cover. A dog he played with last month now scares him. Lasts 2-3 weeks per episode.
 - Management: more space, less pressure, more food. *Don't* push him through. *Don't* drag him forward. Distance is your friend.
-

Week 37-39 (Nov 2-22) — 9 Months

Body

- 34-48 lb (15.4-21.8 kg); ~1095-1345 kcal

Brain

- Beginning to come out of the trough. Trainability slowly returns.

Training goals

- Sport intro: pick one. Agility foundations (jump grids, low cavalettis, body awareness — NO jumping until growth plates close), nosework, treibball, disc, scent work. He needs a *job*.
 - Add long-line recall in DII-7 environments (busier park, dog in distance).
-

PHASE 5 — Late Adolescence & Settling (Weeks 40-52)

Week 40-43 (Nov 23 - Dec 20) — 10 Months

Body

- 35-50 lb (15.9-22.7 kg); ~1110-1360 kcal
- Growth has slowed dramatically. Final ~10% of adult weight comes on slowly through week 52.

Brain

- Prefrontal cortex maturing. Impulse control becoming *available* (not yet reliable).

Training goals

- First off-leash work in fully enclosed spaces (friend's fenced yard, training facility). Long line still attached.
 - Begin building duration: 5-min settle, 30-sec sit-stay at distance.
-

Week 44-47 (Dec 21 - Jan 17) – 11 Months

Body

- 35-52 lb; ~1115-1370 kcal

Brain

- Calmer. The 9-month dog and the 11-month dog often look like different animals.

Training goals

- Generalization push. Take everything he knows and run it in 5 new environments per week.

Vet

- Annual booster + heartworm test typically scheduled around 1 year.
-

Week 48-52 (Jan 18 - Feb 15) – Approaching 12 Months

Body

- 35-55 lb (15.9-25 kg); transitioning to maintenance calories
- *Around now, transition to adult food gradually over 7-10 days.* Calorie multiplier drops to $\times 1.6$ of RER.

Brain

- Brain ~90-95% of adult mature wiring. Personality is locked in (mostly).
- Continued myelination of prefrontal cortex through ~18-24 months.

Training goals

- Spay/neuter decision: now is the *earliest* reasonable point for medium herding breed. Most current research supports waiting until 18 months for males, after first heat for females.
- First adult sport class if not already enrolled.

Vet

- 1-year wellness [done]

- Adult-food transition [done]
 - Heartworm test [done]
-

YEAR ONE COMPLETE

What you have at 12 months: a dog who has experienced ~50 weeks of intentional reinforcement, knows ~15-25 trained behaviors well, has been positively exposed to >150 environments, and has built a relationship that will hold through the second adolescent dip (12-18 months for medium breeds).

The work isn't done. Adolescence runs to ~18-24 months in medium breeds. But the foundation is laid.

QUICK REFERENCE: TYPICAL BC GROWTH CURVE

Age	Female (lb)	Male (lb)	Notes
3 mo	8-13	10-15	Currently here
4 mo	12-18	15-22	Multiplier shifts to ×2
5 mo	16-22	19-27	Adolescence may begin
6 mo	18-26	22-30	Adolescence officially
7 mo	21-29	26-34	
8 mo	24-32	29-37	Peer-reviewed trainability trough
9 mo	26-34	31-40	Sport intro
10 mo	28-37	33-43	Off-leash work begins
11 mo	29-38	34-45	
12 mo	30-40	35-48	Transition to adult food
18 mo	30-45	35-55	Full adult weight

Ranges are wide because BC genetics are wide. Working lines run small and lean; show lines run bigger. Body condition is more important than the number on the scale.

Chapter 2 • The Neuroscience of Puppyhood

The point of this document: when you understand *why* Bucco does what he does at each stage, the training plan stops feeling arbitrary. The behavior follows from the biology.

1. The Critical Period: A Time-Limited Wiring Window

The foundational work here is Scott & Fuller's *Genetics and the Social Behavior of the Dog* (1965), conducted at the Jackson Laboratory in Bar Harbor, Maine. Over 20 years they raised hundreds of puppies under controlled conditions and isolated when, in development, specific kinds of learning are easiest or hardest. Their core finding still holds:

There is a window from roughly 3 to 12-14 weeks during which the puppy brain is uniquely plastic for forming social attachments and accepting novel stimuli as “safe.” After it closes, the same exposures form much more slowly — and often incompletely.

Modern work (Howell et al., Serpell & Duffy, the 2022 *Animals* narrative review) refines this:

- The window opens around week 3, when eyes and ears come online.
- Peak plasticity is roughly **weeks 4-12**.
- Closure is gradual, typically by **weeks 12-16**.

What's happening biologically: the visual cortex builds a template for “what is a social partner” (Luescher's review, Avidog). Until ~12 weeks, this template is wide — it can include cats, parrots, mailmen, vacuum cleaners. After ~14 weeks, the template solidifies. Things outside it are treated as potential threats.

This is why Bucco being in his apartment with two cats from day one is huge. Kiki and Rainier got into the social template before it closed. He'll have a Different Relationship with cats than a BC raised without them.

This is also why what doesn't get in now may not get in later. Vacuums he never met. Men in helmets he never met. Skateboards. Wheelchairs. Children in winter coats. After ~14 weeks, each of these is a *novel* stimulus the adult brain processes as potentially dangerous.

The Scott-Fuller “35/65” finding

Genetics ≈ 35% of adult behavior. Environment (socialization, nutrition, health, training) ≈ 65%. This number is approximate and breed-dependent, but the directional finding holds across decades of replication: **upbringing matters more than breeding for most behavioral traits**. The exception: breed-typical motor patterns like the BC's eye-stalk-chase, which are highly heritable (see §bc-genetics).

2. The Two Fear Periods

Two distinct windows when the puppy's brain becomes temporarily extra-sensitive to scary experiences. The brain is *trying* to learn what's dangerous, and it's faster at that learning than at any other point.

First Fear Period: ~8-11 weeks

Coincides with the tail end of the primary socialization window. Bucco is in this period right now (he turned 12 weeks on May 11).

Mechanism: the amygdala, which processes threat, is fully online before the prefrontal cortex (which evaluates whether the threat is real). A scary event isn't filtered through "is this really dangerous?" — it goes straight into emotional memory.

Consequence: a single bad experience now can produce a lifetime phobia.

Implication: the things Scott does to *create distance* and *feed proactively* are not coddling. They're preventing the amygdala from imprinting a wrong fear.

Second Fear Period: 6-14 months (typically 8-11 months for medium breeds)

Lasts 2-3 weeks at a time, may cycle. Driven partly by hormonal surges (testosterone, estrogen) and partly by the asymmetric brain development of adolescence — emotional brain ahead of executive brain.

Look for: sudden fear of things he was fine with last week. A manhole cover. A specific person. A doorway. It's not a regression in training; it's a temporary recalibration of his threat assessment.

Manage with: distance, food, time. Don't force exposure. Don't punish the fear ("there's nothing to be afraid of!" is meaningless and frustrating). Wait it out at a comfortable distance, feed, leave on a positive note.

Source: AKC, Companion Animal Psychology, multiple veterinary behaviorists.

3. Sleep, REM, and the Puppy Brain

Adult dogs sleep ~12-14 hr/day. Puppies under 4 months sleep **18-20 hr/day**. Bucco needs this much sleep right now.

Why so much? Two reasons:

Memory consolidation

REM sleep is when the day's experiences get filed away. A 2021 study (Iotchev et al., *Scientific Reports*, "Developmental features of sleep electrophysiology in family dogs") tracked sleep across 2-14 months and found:

- Puppies under 6 months spend **~30% of total sleep in REM** (vs. ~10% in adults)
- REM declines steadily from week 8 to week 24
- The decline parallels increasing behavioral complexity — they need less REM as their day-to-day learning load stabilizes

For Bucco: every minute he sleeps is a minute his brain is replaying and consolidating what he learned during his 5 minutes of Crate Games. **Cutting sleep cuts learning.**

Brain growth

Brain weight doubles between weeks 3-12. Synaptic density peaks around weeks 6-8 then gets pruned through the first year — keeping useful connections, discarding unused ones. REM sleep is one of the windows where this pruning happens.

Practical implication. When Bucco is wired, biting harder, ignoring cues, fixating — first ask: how much has he slept today? Hyperactivity in BCs is almost always a sleep symptom, not an exercise deficit. The cure is more crate time, not more walking.

4. Myelination & Impulse Control

Myelination is the process of insulating neurons with a fatty sheath (myelin) so they transmit signals fast and reliably. Different brain regions myelinate on different schedules:

Region	Function	Myelination matures
Sensory cortex (vision, hearing)	Sensory processing	Mostly by 6-8 weeks
Motor cortex	Movement	By 4-5 months
Limbic system (amygdala etc.)	Emotion	Functionally online by 8 weeks
Hippocampus	Memory	Mostly by 6 months
Prefrontal cortex	Impulse control, decision-making	18-24 months

This is why **Bucco's prefrontal cortex is not capable of meaningful impulse control yet, and won't be for another year and a half.** When he grabs the thing he shouldn't, he's not "choosing" to disobey. The wiring that allows him to choose is incomplete. Your job is to manage the environment and reinforce the right answer until the wiring catches up.

This is also why "no" doesn't work the way owners want it to. The frontal cortex can't process "no" the way it can process "*do this instead.*" Replacement behavior > correction.

5. Classical Conditioning (Pavlov) — How Bucco Feels

Pavlov's bell. A neutral stimulus (bell), repeatedly paired with a meaningful stimulus (food), takes on the meaning of the meaningful stimulus. The dog ends up salivating at the bell alone.

For Bucco, every pairing he experiences is doing this whether or not you mean it to:

- Crate + food = crate predicts food → crate is good
- Crate + isolation + crying = crate predicts distress → crate is bad
- Vet building + treats + calm exam = vet predicts good → vet is fine
- Vet building + restraint + needle = vet predicts pain → vet is scary forever

This is why **Susan Garrett's emphasis on classical conditioning before operant conditioning is biologically correct.** How Bucco *feels* about a stimulus dictates how he can *act* around it. A dog who feels safe near other dogs can be trained to ignore them. A dog who feels threatened near other dogs cannot — no amount of operant training overcomes the underlying classical association.

The “+CER” (Conditioned Emotional Response, positive) approach: every new stimulus is introduced at distance, paired with food, until the puppy’s emotional response to it is positive. Then operant training can happen on top of that foundation.

6. Operant Conditioning (Skinner) — How Bucco Acts

Skinner’s box, simplified: behavior that produces good outcomes increases. Behavior that produces bad outcomes (or no outcomes) decreases.

Four “quadrants” — but Garrett only uses one:

	Add	Remove
Behavior ↑ (Reinforcement)	+R: Add good thing → behavior increases	-R: Remove bad thing → behavior increases
Behavior ↓ (Punishment)	+P: Add bad thing → behavior decreases	-P: Remove good thing → behavior decreases

+R only. Add a reinforcer (food, play, environmental access) when the behavior you want happens. Ignore or manage the behavior you don’t want. This is “positive reinforcement training.”

Why this works neurologically: dopamine is released *in anticipation* of reward, not just on receipt. A dog trained with +R produces dopamine at the cue, again at the behavior, again at the marker, again at the food. The whole sequence is rewarding to perform.

Punishment-based training instead produces cortisol (stress hormone). Cortisol disrupts memory consolidation, increases anxiety, and damages the relationship that makes future training possible. There’s no neurochemical advantage to it.

Reinforcement schedules

- **Continuous (CRF):** every correct response gets reinforced. Best for *teaching* a new behavior. High learning rate, but the behavior weakens fast if reinforcement stops (“extinction”).
- **Fixed Ratio (FR):** reinforce every Nth response. Predictable, OK for maintenance.
- **Variable Ratio (VR):** reinforce on average every Nth response, but unpredictable. **Strongest behavior, most resistant to extinction.** This is the slot-machine schedule — the same mechanism that drives gambling addiction is what makes a well-trained recall solid.

For Bucco right now: continuous reinforcement during learning. Variable ratio only once the behavior is fluent — and “fluent” means 99% reliable in that specific context.

The Premack Principle

David Premack, 1965: a higher-probability behavior reinforces a lower-probability behavior. Translation: “do this less-fun thing, then you can do that more-fun thing.”

For Bucco:

- Sit *then* the door opens (loose-leash walks start here)

- Settle on mat *then* greeting Nina when she comes home
- Drop the toy *then* tug restarts
- Look at me *then* you can go sniff that spot

This is one of the most underused tools in pet dog training, and one of the most powerful — because the reinforcer is *built into the environment*. No treat pouch needed.

7. Markers: Why “Yes!” or a Click Matters

The marker problem: by the time Scott reaches for the treat pouch and produces a treat, ~2 seconds have passed. The behavior that just got reinforced isn’t necessarily the one Scott wanted.

The marker (a click, the word “yes!”) solves this by acting as a *bridge stimulus*. Pavlovian-conditioned to reliably predict food, the marker tells Bucco “the thing you just did is the thing I’m paying for,” even if the food itself takes 2 seconds to arrive.

This compresses the learning timeline dramatically. Karen Pryor, drawing from her dolphin training work in the 1960s (animals you can’t physically reach to deliver food to immediately), formalized the technique into “clicker training.” Research (IAABC Foundation Journal review) shows marker training accelerates acquisition of complex behaviors by ~50% vs. food alone, though for simple behaviors the advantage is smaller.

Timing matters. The marker needs to fire within ~0.5 seconds of the behavior. Beyond ~1.5 seconds, the association weakens.

Verbal “yes!” vs. clicker: clicker is more precise (it’s the same sound every time, unaffected by Scott’s tone or mood). Verbal is more available (Scott always has his voice). For Bucco, verbal is fine as long as Scott picks one short, distinct word and uses it consistently.

8. The Gut-Brain Axis (Newer Science)

Recent research has shown a real connection between gut microbiome composition and behavior. The vagus nerve carries signals from the gut to the brain; the gut microbiome produces or modulates several neurotransmitters (GABA, serotonin, dopamine) directly. A 2024 review (Saco et al., *Veterinary Medicine International*) and a 2025 *Scientific Reports* paper found:

- The gut bacterial genus *Blautia* is consistently linked to anxiety scores in companion dogs
- The probiotic strain *Bifidobacterium longum* NCC3001 reduced anxious behaviors in a controlled trial
- *Lactiplantibacillus plantarum* PS128 improved aggression and separation-anxiety scores after 14 days

This is emerging science. Don’t treat probiotics as a fix for behavioral problems. But:

- Feeding consistency matters more than the brand of kibble
- Avoid frequent antibiotic courses unless medically necessary
- A high-quality food with prebiotic fibers supports a healthier microbiome
- Frequent diarrhea = vet visit, not just a hygiene problem — chronic GI disruption can affect behavior

9. Stress, Cortisol, and the Three-Day Rule

Acute stress (a single scary event) raises cortisol for hours. Chronic stress raises baseline cortisol, which:

- Impairs memory consolidation (so chronically-stressed dogs learn slowly)
- Increases reactivity (lower threshold for fear/aggression responses)
- Disrupts sleep
- Compromises immune function

The 72-hour rule: after a significantly stressful event (a fight, a vet emergency, fireworks, a fall), cortisol takes ~72 hours to return to baseline. *Don't train hard for three days after.* Don't expose to triggers. Let the system reset.

For Bucco: post-vaccine days, the day after a fear-inducing event, the day after a long car trip — these are recovery days, not training days. Slow walks, sniffing, soft food, quiet.

10. The Practical Takeaway

If you remember nothing else from this document, remember:

1. **The primary socialization window closes around 14 weeks.** Pack everything you can in.
 2. **REM sleep is when learning gets consolidated.** Protect his 18-20 hours.
 3. **Classical conditioning before operant conditioning.** How he feels precedes what he does.
 4. **The prefrontal cortex isn't done myelinating until 18-24 months.** Impulse control is biologically limited.
 5. **Adolescence is real and peer-reviewed.** Trainability dips around 8 months. Don't take it personally.
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Chapter 3 • Feeding Science

What goes in matters more than most owners realize. For a growing medium-breed working dog, “good enough” food + slightly off macros = orthopedic problems at year three. This document walks through what to feed, how much, why, and where the science actually points.

1. Caloric Requirements — The Math, Properly

Resting Energy Requirement (RER)

The baseline metabolic burn. Every dog, regardless of activity:

$$\text{RER (kcal/day)} = 70 \times (\text{body weight in kg})^{0.75}$$

This is the Kleiber equation, the empirical metabolic-scaling law that holds across mammals from mice to elephants. The exponent (0.75) reflects that smaller bodies burn proportionally more per gram than larger ones.

Maintenance Energy Requirement (MER)

RER multiplied by a lifestyle factor:

Lifestyle/Stage	Multiplier	Notes
Puppy 0-4 months	×3.0	Highest of any life stage
Puppy 4-12 months	×2.0	Growth slows
Adult intact, active	×1.6	Working BCs often run here
Adult intact, moderate	×1.4	Pet dog standard
Adult neutered, light	×1.2-1.4	Neutering reduces metabolism ~25%
Senior (7+)	×1.2	
Working dog (active)	×2-5	Sled dogs hit ×8 in winter

Bucco's current target (week 12, ~12-18 lb)

Weight	kg	RER	×3 (kcal/day)
12 lb	5.4	248	~745
14 lb	6.4	282	~845
16 lb	7.3	312	~935
18 lb	8.2	343	~1030

When the multiplier drops to ×2 (around 4-5 months)

Weight	kg	RER	×2 (kcal/day)
20 lb	9.1	369	~740
25 lb	11.3	437	~875

Weight	kg	RER	×2 (kcal/day)
30 lb	13.6	501	~1000
35 lb	15.9	564	~1130
40 lb	18.1	622	~1245

You'll notice calorie target stops climbing rapidly once the multiplier drops. This is correct. **Growth slows; caloric need plateaus; weight continues climbing on the same calorie budget because activity increases too.**

2. Macronutrients – What the Calories Should Be

Per AAFCO (Association of American Feed Control Officials) puppy growth standards, dry matter basis:

Nutrient	Minimum	Notes
Protein	22.5%	Quality matters: animal-based > plant-based amino acid profiles
Fat	8.5%	Critical for brain development; DHA omega-3 is the big one
Calcium	1.0%	Maximum 1.8% for large-breed growth
Phosphorus Ca:P ratio	0.8% 1.1:1 to 1.3:1	This ratio matters more than absolute calcium

Protein: How much, what kind?

Old myth: “high protein causes orthopedic problems.” Demolished by Nap & Hazewinkel’s 1991 study in the *Journal of Nutrition* – Great Dane puppies fed 14.6%, 23.1%, or 31.6% protein diets showed **no differences in skeletal development**. Protein was never the culprit.

What does protein quality mean? Amino acid profile. Dogs require 10 essential amino acids: arginine, histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, valine. Animal proteins (meat, eggs, dairy) have all 10 in good ratios. Plant proteins are usually deficient in 1-3 – so a chicken-and-rice food generally beats a pea-and-rice food on amino acid availability.

Fat: The brain fuel

Brain dry matter is ~60% fat. The growing puppy brain needs DHA (docosahexaenoic acid, a long-chain omega-3) for membrane and myelin formation. Studies (Zicker et al., Hill’s research) have shown puppies fed DHA-enhanced diets score better on visual and cognitive learning tasks.

Look for: fish oil, salmon, algae oil in the ingredients. Minimum DHA: 0.05% dry matter for growing puppies.

Carbs: Not the enemy, not required

Dogs have no minimum carbohydrate requirement. They can derive glucose from protein via gluconeogenesis. **But** carbs are a cheap energy source and most kibbles use them. There's nothing wrong with a moderate-carb diet for a non-working pet dog. Grain-inclusive vs. grain-free is largely marketing (see DCM discussion below).

3. Calcium, Phosphorus, and The Single Most Important Number

This is the most important macro for a large/medium-breed puppy. Get it wrong and you're looking at hip dysplasia, OCD (osteochondrosis), angular limb deformities, or compression fractures.

Hazewinkel's seminal 1985-1991 work at Utrecht demonstrated:

- Excess dietary calcium in growing large-breed puppies causes osteochondrosis, delayed endochondral ossification, and angular limb deformities — **independent of total energy intake.**
- A 2017 case report (PMC 5731398) documented a large-breed puppy with compression fractures, vitamin D deficiency, and suspected nutritional secondary hyperparathyroidism from an unbalanced homemade diet.

What to look for on the bag

- **AAFCO statement:** must say "complete and balanced for growth" or "for all life stages including growth of large size dogs (70 lb or more as an adult)." For Bucco (medium breed, ~35-50 lb adult), the "growth" statement is sufficient — he's not large enough to require the large-breed-specific statement, but a food formulated for it is fine.
- **Calcium:** 1.0-1.8% dry matter for growing puppies. Lower end (1.0-1.4%) is safer for medium/large.
- **Phosphorus:** 0.8-1.6%.
- **Ca:P ratio:** 1.1:1 to 1.3:1 is ideal. Above 1.6:1 or below 1.0:1 is concerning.

What NOT to do

- Don't supplement calcium "to help his bones grow." This is the most common harmful mistake. The calcium in his food is what his bones need; more is harmful, not better.
 - Don't feed adult dog food during growth. Adult formulas don't guarantee the calcium/phosphorus profile he needs.
 - Don't switch foods every month. Sudden GI disruption + nutrient profile changes during growth = trouble.
-

4. The Grain-Free / DCM Story

In 2018, the FDA opened an investigation into a possible link between grain-free diets (specifically those high in peas, lentils, legumes, potatoes) and **dilated cardiomyopathy (DCM)** in breeds not historically prone to it.

State of the science as of late 2024 / 2025:

- **No causative link has been definitively established.** The FDA's investigation found a statistical correlation but couldn't isolate a single mechanism.
- One hypothesis: taurine deficiency related to the legume-heavy formulations. Some affected dogs had low blood taurine; many didn't.
- Many initially-affected dogs improved on a diet change, suggesting *something* dietary is involved.
- The FDA narrowed its focus to specific brand families with high legume content and "boutique" formulations from smaller manufacturers.

Practical recommendation: Without a documented grain allergy or sensitivity, default to a grain-inclusive food from a manufacturer with veterinary nutritionists on staff and a long track record. Brands meeting the "WSAVA Global Nutrition Committee" recommended criteria include Royal Canin, Hill's, Purina Pro Plan, and Eukanuba. (This is not an endorsement; it's a risk-reduction default.)

If Bucco has confirmed grain allergy: grain-free is fine, but pick one with limited legume content and animal protein as #1 ingredient.

5. Meal Frequency

Age	Meals per day	Reason
8-16 weeks	4	Small stomach, fast metabolism, hypoglycemia risk
4-6 months	3	Stomach capacity increases
6-12 months	2-3	Growth slowing
12+ months	2	Adult maintenance

Why not free-feed? Three reasons:

1. *Lost training currency.* If food is always available, food is no longer reinforcing.
2. *Obscured appetite as health signal.* A skipped meal is one of the earliest signs of illness; you won't notice if you don't track.
3. *Resource-guarding prevention.* Food has structure and ritual; dogs don't feel scarce around it.

6. Treats as a Training Tool

Hierarchy of value (Bucco-specific, build with his preferences)

- **Tier 1 (lowest):** his regular kibble. Sufficient at home, low distraction.
- **Tier 2:** training treats, freeze-dried liver, small cheese cubes, baby carrots.
- **Tier 3:** boiled chicken breast, hot dog pieces (sodium! sparingly), the occasional small piece of *plain* scrambled egg.
- **Tier 4 (rare):** something he loses his mind over. For Bucco, this might be a specific cheese, real liver, or a high-value tug toy.

Match value to difficulty. Recall in the backyard? Kibble might do. Recall away from a squirrel? Tier 3-4.

The 10% rule (or 20% if using kibble)

If treats are *separate* from his daily food: total treat calories $\leq$ 10% of daily calories.

If you're using his daily kibble *as* training treats (the Garrett default at home): up to ~80% of his daily food can be delivered through training. The total stays the same; the delivery method shifts.

Treat math example

Bucco at 15 lb, week 12, 880 kcal/day target.

- 10% treat cap (if separate): ≤ 88 kcal in treats.
- Typical training treat = 2-3 kcal. That's ~30-40 small treats *of separate treats* per day.
- If kibble: $\sim \frac{1}{2}$ cup kibble = ~ 200 kcal = ~ 100 -150 individual pieces of kibble.

What NOT to give

Toxic to dogs, full stop:

- **Grapes, raisins** — acute kidney failure, mechanism still poorly understood
- **Onions, garlic, leeks** — hemolytic anemia
- **Chocolate** — theobromine and caffeine toxicity (darker = worse)
- **Xylitol** — sweetener in sugar-free gum, peanut butter, mints; causes life-threatening hypoglycemia
- **Macadamia nuts** — neurological symptoms
- **Alcohol** — fatal
- **Cooked bones** — splinter, perforate GI tract
- **Avocado** — persin, mostly an issue in large quantities
- **Yeast dough** — ferments in stomach
- **Apple cores / stone fruit pits** — cyanide compounds

7. The Body Condition Score (BCS) — The Real Metric

Numbers on a scale are imprecise (a kibble swing, a recent meal, a recent pee all shift it). BCS is the standard.

9-point scale (Purina / WSAVA):

Score	Description
1	Emaciated. Ribs, vertebrae, pelvic bones obvious from a distance. No fat.
2	Very thin. Bones clearly visible, minimal flesh.
3	Thin. Ribs easily palpable, slight fat covering. Tops of vertebrae visible.
4	Underweight. Ribs easily palpable with minimal fat. Visible waist.
5	Ideal. Ribs palpable without excess fat. Waist visible from above. Tummy tuck from side.

Score	Description
6	Slightly overweight but acceptable for many BCs.
7	Overweight. Ribs palpable with effort, fat covering. Waist hard to see.
8	Obese. Ribs hard to find. Pendulous belly. No waist.
9	Severely obese. Massive fat deposits.

For a growing BC, target BCS 4-5. Lean wins for joint health and longevity (Purina's landmark 14-year lifespan study on Labradors found lean dogs lived ~1.8 years longer on average).

Recheck every 2 weeks during growth. Adjust by $\pm 10\%$ on the calorie target.

8. Water

A growing puppy needs $\sim \frac{1}{2}$ to 1 oz per pound of body weight per day. For Bucco at 15 lb: roughly 7-15 oz/day. More on hot days, more after exercise.

Provide fresh water at all times. Bowl: stainless steel or ceramic (plastic harbors bacteria, can cause chin acne). Wash daily.

9. Supplements: What Bucco Actually Needs

If you're feeding a complete and balanced food meeting AAFCO growth standards, **the answer is almost always: nothing.** But:

- **Fish oil** — extra omega-3 (DHA/EPA) for brain development and skin. 1000mg fish oil per ~10 lb body weight, split daily. Pick a brand with third-party testing for mercury and PCBs.
- **Probiotics** — *if* GI issues are recurring. Otherwise marginal. Specific strains with research: Bifidobacterium longum, Lactobacillus plantarum.
- **Joint supplements (glucosamine, chondroitin)** — research is mixed for healthy young dogs. Most veterinary nutritionists don't recommend in puppyhood.
- **NEVER** supplement calcium, vitamin D, or vitamin A. All three can cause harm in excess.

10. Food Transitions

Every change should happen over 7-10 days. Sudden changes = osmotic diarrhea (water rushes into the gut to dilute the unfamiliar contents).

Day	Old food	New food
1-2	75%	25%

Day	Old food	New food
3-4	50%	50%
5-6	25%	75%
7+	0%	100%

Watch stool quality. Slight loosening for 2-3 days is normal. Frank diarrhea, vomiting, or blood = slow the transition or back off entirely.

11. The Microbiome Angle

Newer research (covered in \$neuroscience) links gut microbiome composition to canine anxiety and aggression. The actionable items for Bucco:

- **Feeding consistency over feeding novelty.** Frequent food changes disrupt the microbiome.
- **Limit antibiotic exposure** to when medically required.
- **Prebiotic fibers in the food** support beneficial gut bacteria. Look for: chicory root, beet pulp, FOS (fructooligosaccharides).
- **A spoon of plain unsweetened yogurt or kefir** a few times a week is well-tolerated by most dogs and provides probiotic strains. *Skip if any lactose sensitivity.*

12. The Susan Garrett “Train With Meals” Philosophy – Why It’s Sound

Bucco gets ~80% of his daily food *through training*, not from a bowl.

Why this works:

1. **Engagement.** Food from the handler builds the handler as the source of good things.
2. **Reinforcement budget.** A puppy eats 4× per day. Each meal becomes 100+ training reps. That’s 400+ reps/day at no additional caloric cost.
3. **Calm meal patterns.** Eating quietly from a hand, a mat, a Kong, or a snuffle mat trains arousal regulation around food.
4. **Default behaviors.** “When food appears, what do you offer?” becomes a question the puppy starts answering proactively.

What the remaining ~20% looks like: frozen Kongs, lick mats, snuffle mats, food-dispensing toys — all enrichment, all building independent calm play.

13. Quick Reference Card

CURRENT (week 12, est. 15 lb)

- 880 kcal/day
- 4 meals
- BCS target: 4-5
- ~80% through training, ~20% enrichment

ADJUSTMENTS

- Too thin (BCS ≤ 3): +10%, recheck 2 wk
- Too heavy (BCS ≥ 6): -10%, recheck 2 wk
- Sluggish, loose stool: investigate; consider vet
- Refusing food >24 hr: vet now

CA:P CHECK ON BAG

- Ca: 1.0-1.4% (medium breed)
- Ca:P ratio: 1.1-1.3 : 1
- AAFCO statement: "complete & balanced for growth"

NEVER

- Adult food during growth
 - Calcium / vitamin D / vitamin A supplements
 - Sudden food changes
 - Cooked bones
 - Grapes, raisins, onions, garlic, chocolate, xylitol, macadamias
-

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Chapter 4 • Learning Theory

The Garrett framework isn't a brand. It's a clean application of behavioral and learning science assembled by someone who's spent four decades testing it in agility, the most consequence-dense dog sport that exists. This document unpacks the science under each piece — why high rate of reinforcement works, why "no" doesn't, why classical conditioning has to come first.

1. The Four Building Blocks

All animal learning sits on four mechanisms. Anyone training a dog is using these whether they know it or not.

1.1 Habituation

Reduced response to a repeated stimulus that turns out to be neither rewarding nor harmful. The doorbell rings 100 times with nothing scary attached → eventually the puppy stops reacting.

Bucco implication: "ignore the doorbell" gets trained by feeding kibble through 50 reps of the doorbell ringing while nothing important happens. Not by yelling "QUIET."

1.2 Sensitization

Increased response to a repeated stimulus that's been paired with negative outcomes. The doorbell rings → the resident dog bursts in barking → the puppy associates doorbell with chaos → over time the puppy reacts harder to the doorbell, not softer.

Bucco implication: habituation requires neutral repetitions. If the puppy is reacting and you're not managing it, you're sensitizing him, not habituating him.

1.3 Classical Conditioning (Pavlov, 1903)

A neutral stimulus paired repeatedly with a meaningful stimulus takes on the meaning. Bell + food → bell alone produces salivation.

For Bucco, every pairing is doing this: - Crate + food + sleep + calm = crate is good → he runs in willingly - Crate + isolation + crying = crate is bad → he resists the crate

The biology: the amygdala (emotion) and hippocampus (memory of stimuli) form a fast, durable link. Once a strong classical association is formed, it can be modified but not erased.

1.4 Operant Conditioning (Skinner, 1938)

Behavior is shaped by its consequences. Behaviors that produce rewards increase; behaviors that produce nothing (or punishment) decrease.

The biology: the basal ganglia and dopamine system reinforce action-outcome pairings. Dopamine isn't released when the reward arrives — it's released in anticipation, when the cue or behavior predicts reward. This is why the *expectation* of reward is what drives behavior.

2. The Operant Quadrants — And Why Garrett Only Uses One

Stimulus	Behavior increases	Behavior decreases
Added	+R (positive reinforcement)	+P (positive punishment)
Removed	-R (negative reinforcement)	-P (negative punishment)

- **+R**: add a treat after sit → sit increases. THE QUADRANT.
- **+P**: leash pop after pulling → pulling decreases. Damages the relationship; creates fear of the handler.
- **-R**: ear-pinch until dog picks up dumbbell → dog learns to pick up dumbbell to make the pinch stop. Used in some traditional methods. Cruel and unnecessary.
- **-P**: turn away when dog jumps → jumping decreases. Useful adjunct to +R for some attention-seeking behaviors. NOT a primary tool.

Why +R-only?

1. **Neurochemistry.** +R produces dopamine. +P and -R produce cortisol. Cortisol disrupts learning. Dopamine accelerates it.
2. **Relationship.** The handler becomes a predictor of good things, not bad. This is the relationship that holds in adolescence.
3. **Generalization.** +R-trained behaviors transfer between contexts better. +P-trained behaviors collapse outside the original context (because they're context-bound — the dog learned “stop here or it hurts,” not “this is wrong”).
4. **No collateral damage.** +P can suppress one behavior and cause five unrelated fears or aggressions to emerge. Documented across studies; see Ziv 2017 review in *Journal of Veterinary Behavior*.

3. Reinforcement Schedules — Why VR Wins

Continuous (CRF)

Every correct response is reinforced. **Use for:** teaching a new behavior. **Drawback:** weak resistance to extinction. Stop reinforcing and the behavior fades fast.

Fixed Ratio (FR)

Every Nth correct response is reinforced. **Use for:** maintenance. **Drawback:** “post-reinforcement pause” — the dog learns the pattern and slows down right after a reward.

Fixed Interval (FI)

Reinforcement available every N seconds, regardless of how many responses. Produces a “scallop” pattern — responding peaks just before each interval ends.

Variable Ratio (VR)

Reinforcement on average every Nth response, but unpredictable. **The strongest schedule.** Highest response rate. Most resistant to extinction. This is the slot-machine schedule.

Variable Interval (VI)

Reinforcement on average every N seconds, but unpredictable. Moderate response rate; very durable behavior.

For Bucco — practical schedule progression:

Stage 1 (teaching): CRF – every rep gets food
Stage 2 (early fluency): CRF still, but small chunks of variable food (1 piece vs 3)
Stage 3 (fluency): VR-2, then VR-3 (random pattern)
Stage 4 (maintenance): VR-5 to VR-10, plus environmental reinforcers (Premack)
Stage 5 (life): Occasional jackpot reinforcement; mostly environmental

Never go to a fixed schedule. It creates pauses. Always keep some variability.

4. Rate of Reinforcement (RoR) — The Single Most Underrated Variable

When teaching a new behavior, Garrett's guidance is **15-30 reinforcements per minute**. Three to five seconds between markers.

Why this matters:

- **Attention is preserved by frequency.** If reinforcement is rare, the dog disengages between deliveries. If frequent, the dog stays plugged in.
- **Working memory window is short.** The puppy must connect the behavior to the reward; large gaps make this fail.
- **Dopamine system stays engaged.** A high pulse rate keeps the predictive expectation high.

Compare: - **Low RoR (2-3/min):** "Sit." (wait) "Good boy." (wait) "Stay." (wait) "OK release!" → boring, low engagement, slow learning. - **High RoR (20+/min):** Capture sit → marker → food → next rep within 3 seconds → marker → food → marker → food → end on a win → walk away. Three minutes of this produces more learning than 30 minutes of low-RoR work.

5. Shaping — Building Behaviors That Don't Exist Yet

You can't reinforce a behavior that doesn't happen. So you reinforce *approximations* and gradually raise the bar.

Example: shaping a "spin" cue from nothing.

1. Dog looks slightly left → mark + treat
2. Dog turns head left → mark + treat (you stopped reinforcing the head look)
3. Dog moves left front paw → mark + treat

4. Dog rotates body 1/8 turn left → mark + treat
5. Dog rotates 1/4 turn → mark + treat
6. Dog rotates 1/2 turn → mark + treat
7. Dog completes full spin → mark + treat
8. Add the verbal cue “spin”

Each step is a *successive approximation* of the final behavior. The skill of the trainer: break the behavior into small enough steps that the dog succeeds at each one. Too big a jump → dog fails → frustration → quits.

Common shaping behaviors for Bucco

- **Hand target:** reinforce looking at the hand → sniffing the hand → nose-touching the hand.
- **Settle on mat:** reinforce walking near the mat → stepping on it → lying on it → staying for 1 sec → 2 sec → etc.
- **Crate Games:** look at crate → step toward it → step in → all four feet in → turn around → lie down → stay 1 sec → etc.

Errorless learning

Garrett emphasis: set the environment so the puppy succeeds. Don't lure for too many reps (he becomes lure-dependent). Don't ask too much too soon (he fails and learns frustration). When a session goes badly, the failure is on the trainer for setting too big a step.

6. Markers – The Bridging Stimulus

Karen Pryor's clicker work, originating in dolphin training in the 1960s, formalized the marker. A short, distinct sound delivered the instant the correct behavior happens, followed by food within 1-2 seconds.

Why the marker matters

- The behavior-to-reward gap is the learning bottleneck. By the time food appears, the dog may have moved, scratched, looked away.
- The marker fires within 0.3 seconds of the behavior, anchoring *exactly* what is being reinforced.
- Pavlovian conditioning makes the marker itself rewarding (paired enough times with food, the marker produces a mini-dopamine pulse on its own).

Timing precision

- Within **0.5 sec** of the behavior: ideal
- Within **1.5 sec**: still effective
- Beyond **2 sec**: association weakens; you may be reinforcing whatever the dog did at the 2-sec mark

Clicker vs. verbal “yes!”

	Clicker	Verbal “yes!”
Precision	Identical every time	Varies with mood
Availability	Need to be holding it	Always available
Distinctness	Stands out from environment	Can blend with other talk
Best for	Detail work, shaping, sport	Pet dog daily training

For Bucco: verbal is fine. Pick one short, distinct word (one syllable, sharp consonant). Don’t dilute it by using it casually. The marker should mean *food is coming, right now*.

7. The Premack Principle – Using Life as Reinforcement

David Premack, 1959-1965: a higher-probability behavior reinforces a lower-probability one.

Translation for dogs: **a thing the dog wants to do can be the reward for doing a thing you want.**

Reward (high-probability)	Requirement (low-probability)
Sniffing	Loose leash to the bush
Greeting a person	Sit politely first
Going outside	Wait at the door
Eating dinner	Settle on mat
Tug game restarts	“Drop”
Off-leash freedom	Reliable recall on the long line

Why it’s underused

Owners default to food because it’s easy. But food gets boring; food gets satiated. Environmental rewards never satiate — the dog *always* wants to sniff that bush.

For Bucco — apply Premack everywhere:

- Don’t open the front door until he’s sitting.
- Don’t put down the food bowl until he’s calm.
- Don’t release the tug toy back into play until he drops on cue.
- Don’t let him sniff that fence until he gives you a 2-second eye contact.

These are not “rules to be strict.” They’re free reinforcers built into the environment.

8. Classical Counterconditioning – The Most Useful Tool For Fearful Dogs

If a dog already has a negative classical association with something (vet, dropped pan, men in hats, vacuum), you can rebuild it. Not erase — replace it with a stronger positive association.

The technique: present the trigger at *sub-threshold distance/intensity* + immediately produce extraordinary reinforcement. Repeat over many sessions. Slowly close the distance / raise the intensity. The dog's emotional response shifts.

Critical: the trigger always appears *before* the food. Trigger predicts food. If you flip the order (food then trigger), you accidentally signal that food predicts the scary thing → food becomes scary.

For Bucco: this is your tool if anything ever scares him. Vacuum, fireworks, a specific person – same protocol every time. Distance, food, repetition, patience.

9. The Distraction Intensity Index (DII) – Generalization Made Concrete

The single biggest training mistake: training a behavior at home, asking for it in the park, and getting frustrated when the dog “doesn’t know it.”

He doesn’t know it. He knows it *at home*. Same behavior, new environment, new internal state – for the dog, it’s almost a different cue.

Garrett’s distraction scale (1-10):

DII	Environment example	When to train here
1	Living room, alone, no toys	Teaching a new behavior
2	Living room, Nina present	Early fluency
3	Hallway, mild distractions	Fluency check
4	Backyard, no other animals	Adding mild challenge
5	Backyard with squirrels visible	Real test of fluency
6	Driveway, occasional passing car	Building generalization
7	Quiet park, no other dogs	Recall begins to be testable
8	Sidewalk, other dogs at distance	Reliability check
9	Busy outdoor patio, kids, dogs	Late-stage proofing
10	Game-day chaos near the stadium	Master level

Rules:

1. Train new things only at DII 1-2.
2. Generalize to a new DII only when 90% reliable at the previous level.
3. If you have to skip a level, go back and fill it in later.
4. **Don’t test what you haven’t trained.** Don’t call your dog off a squirrel if you haven’t trained recall at DII 5-6 first.

10. Why “No” Doesn’t Work

Three biological reasons:

1. **It tells the dog what NOT to do, not what TO do.** The brain doesn’t process negative space well. The dog needs a replacement behavior.

2. **It often triggers cortisol release** even at mild volume. Cortisol disrupts memory of the current event, so the dog doesn't remember what he was just punished for.
3. **It generalizes poorly.** "No" said over a chewed shoe means "don't chew shoes." It doesn't mean "don't chew anything." The dog learns context, not the broader rule.

Better:

- Manage the environment so the wrong behavior can't happen.
- Reinforce the right behavior heavily when it happens.
- Use Premack/IYC so the dog chooses the right thing because it pays better.

11. Extinction – When Behaviors Fade

If a previously-reinforced behavior stops being reinforced, it eventually fades. This sounds simple but two things complicate it:

Extinction burst

The first response to extinction is *more* of the behavior, not less. The dog tries harder. If you don't reinforce a jumping greeting → he jumps harder for a session or two before the jumping subsides.

You must outlast the extinction burst. If you cave during it, you taught the dog that the behavior, when amplified, works.

Spontaneous recovery

A behavior that's been extinguished can pop back up later, often after a break. Treat it the same — don't reinforce, wait for it to fade again.

12. The Garrett Mindset Rules (re-stated, science-anchored)

Rule	Underlying science
"What you grow is what you get."	Operant conditioning — every interaction reinforces <i>something</i> .
Errorless learning.	Successive approximation — small steps prevent frustration, accelerate acquisition.
Recognize vs. know.	Generalization research — context-bound behavior ≠ true cue.
Classical conditioning before operant.	Pavlov + amygdala/hippocampus — emotional state gates operant accessibility.
Premack.	Behavioral economics — relative probability defines what's reinforcing.
No corrections.	Cortisol research — punishment impairs learning and damages relationship.

Rule	Underlying science
High rate of reinforcement.	Attention/dopamine — frequent positive predictions keep system engaged.
End on a win.	Recency effect — last experience anchors emotional memory of session.

13. The Three Questions To Ask Before Any Training Decision

1. **What do I want him to do?** (Define the behavior, not “stop X.”)
2. **What’s reinforcing the unwanted behavior?** (Follow the money — every behavior pays in some currency.)
3. **How do I make the wanted behavior pay better than the alternative?** (Sometimes the answer is environmental management, not training.)

14. When Behaviors Don’t Generalize — The Diagnostic

Bucco won’t sit in the parking lot when he sits at home. What’s actually going on?

Possible cause	Test	Fix
Behavior not actually fluent at home	Ask 10 times at home; succeed ≥ 9 ?	Train at lower DII until fluent
DII too high	Move 20 ft away from distraction	Drop DII, rebuild
Stress / arousal too high	Watch for panting, lip licks, scanning	Decompress first, train later
Hungry/tired/under-slept	Has he eaten? Slept enough?	Reschedule
Reinforcer not valuable enough	Use tier 3-4 treats here	Match treat value to DII
Cue contaminated	Have you been repeating the cue when he doesn’t respond?	Quit using it; re-teach with new cue word

15. The Single Most Useful Habit

Capture, don’t command.

Default offered behaviors are stronger than cued ones, because the dog initiated them. Capturing means: notice the dog doing something good, mark, reinforce, move on. No cue.

- He sits when he sees the food bowl → mark + treat.
- He lies down because he’s tired → mark + treat.

- He looks at you for no reason → mark + treat.
- He passes a cat without staring → mark + treat.

Within weeks, the dog is offering more of these behaviors *spontaneously*. They become defaults. Defaults are what hold up during adolescence.

Sources

- [SimplyPsychology – Operant Conditioning](#)
- [SimplyPsychology – Premack Principle](#)
- [DogNerdly – Premack Principle in Dog Training](#)
- [Whole Dog Journal – Clicker Training 101](#)
- [IAABC Foundation Journal – Clicker training acquisition study](#)
- [PetMD – How to Apply the Premack Principle](#)
- [Study.com – Variable Ratio Schedule](#)

Chapter 5 • Border Collie Genetics & Behavior

The Border Collie isn't a generic dog with extra brain. It's a behavioral specialist — bred for ~150 years (and refined for far longer in the Scottish Borders) to perform one specific neuromotor sequence on command, indefinitely, in harsh weather, over difficult terrain. Every quirk Bucco has makes sense once you understand what he was built to do.

1. Breed History — A Specialist, Not A Generalist

The Border Collie's modern lineage traces to a single dog: **Old Hemp**, a tricolor male born in 1893 in Northumberland on the Scottish-English border. Bred by Adam Telfer, Old Hemp was renowned for an unusually quiet, focused, "stalking" working style — distinct from the more vocal, rougher styles common at the time. He sired ~200 working dogs and is considered the foundation of the modern BC.

The breed wasn't recognized by the AKC until 1995 — over a century after Old Hemp — because the working community resisted formal recognition, fearing show-standard breeding would dilute working ability. They were largely correct. The "split" between show and working lines, which intensified after AKC recognition, produced two functionally different populations:

- **Working line.** Bred for stock-handling ability. Smaller (35-45 lb), often less coat, more variable color/pattern, intense focus, exceptional stamina, lower-key when off-task. Bucco's heritage if his breeder runs sheep.
- **Show/conformation line.** Bred for ring appearance. Larger (45-55+ lb), heavier coat, classic black-and-white markings prized. Often higher arousal off-task because the working focus has been bred away.
- **Sport/performance line.** Hybrid: bred for agility/disc/flyball performance. Drive, tugging, retrieving emphasized. Sometimes lighter on classic stock-handling cues.

Which line is Bucco? Worth asking the breeder. The training approach is the same, but understanding the line helps with realistic expectations for working drive, off-switch ability, and arousal management.

2. The Predatory Motor Pattern Sequence (Coppinger)

Raymond and Lorna Coppinger's foundational work on canine breed behavior identified that all herding behavior derives from a modified wolf predatory sequence:

Orient → Eye → Stalk → Chase → Grab-Bite → Kill-Bite → Dissect → Consume

A wild canid does the full sequence. Domestic dogs, through selective breeding, have had this sequence **broken at different points** depending on what their job is:

Breed type	Pattern emphasized	Pattern suppressed
Livestock guardian (Pyrenees, Anatolian)	None	Entire predatory sequence inhibited
Sighthound (Greyhound)	Chase, grab-bite	Eye, stalk

Breed type	Pattern emphasized	Pattern suppressed
Retriever	Grab-bite (soft)	Kill-bite, consume
Pointer	Eye	Chase, grab-bite
Heeler (cattle dog)	Chase, grab-bite (to heel)	Eye, stalk emphasis
Border Collie	Orient, eye, stalk, chase	Grab-bite, kill-bite, consume

The BC was selected for exaggeration of eye-stalk-chase and inhibition of the terminal grab-bite/kill-bite. This is what makes them the most refined herding dog — they can move stock with eye and body pressure without touching the animal.

Why this matters for Bucco

The neural reward isn't in the kill. It's in the *stalk and chase*. Every staring, freezing, slow-moving moment is its own dopamine pulse. This is why:

- BCs *love* watching things move — squirrels, cars, kids running.
- A BC who chases a cat may not bite the cat — he's after the chase reward, not predation. (But the cat doesn't know that, and chase can still injure or kill.)
- "Eye" behavior (the intense staring) is a self-reinforcing trance. The longer it goes, the harder it is to interrupt.
- Anything that mimics a fleeing prey signature — cars, bikes, joggers, scooters — pulls hard on this circuit.

Management implications

- **Interrupt eye/stalk behavior the moment it starts.** Not "if it escalates." The earlier you redirect, the easier — because the dopamine cascade hasn't started yet.
- **Give him outlets.** A BC denied predatory motor pattern access invents one. Often a pathological one (light chasing, shadow chasing, tail chasing).
- **Sports that channel this:** agility, disc, treibball, even structured fetch. He needs to *do* the motor pattern, just in a structured context.

3. BC Genetic Disorders — What To Know

The breed has a relatively small number of well-characterized genetic diseases. Genetic testing is available for all of these and a responsible breeder will have tested.

3.1 Collie Eye Anomaly (CEA)

- **Frequency:** Affects ~2.5% of Border Collies; carrier rate higher.
- **What it is:** Developmental abnormality of the choroid (the vascular layer behind the retina). Most affected dogs have only mild defects ("choroidal hypoplasia") and live normal lives. Severe forms can cause blindness or retinal detachment.
- **Inheritance:** Autosomal recessive. Both parents must be carriers for offspring to be affected.
- **Test:** OptiGen/UC Davis genetic test, often combined with eye exam by a board-certified veterinary ophthalmologist.

- **Action for Bucco:** Ask the breeder if both parents were CEA-clear or eye-tested.

3.2 Trapped Neutrophil Syndrome (TNS)

- **Frequency:** Carrier rate ~5.9-8% in working and show populations. *Affected* dogs are rare because most breeders test.
- **What it is:** Immune system disorder. Neutrophils (white blood cells) are produced normally but trapped in the bone marrow and don't reach circulation. Affected pups appear normal at birth, then develop infections that don't respond to treatment, usually dying before 4 months.
- **Inheritance:** Autosomal recessive.
- **Action:** If Bucco is healthy at 12 weeks, he's not affected. But knowing his carrier status matters if he's ever bred.

3.3 Neuronal Ceroid Lipofuscinosis (NCL)

- **Frequency:** Mutant allele ~3.5% population frequency. Affected dogs rare.
- **What it is:** A lysosomal storage disease. Lipofuscin (a cellular waste product) accumulates in neurons. Symptoms (anxiety, behavior change, loss of coordination, blindness) emerge between 18-24 months. Always fatal.
- **Inheritance:** Autosomal recessive.
- **Test:** Genomia, UC Davis, others.

3.4 MDR1 (Multi-Drug Resistance)

- **Frequency:** Lower in Border Collies than other collie-type breeds (Australian Shepherds, Collies, Shelties). The MDR1-1Δ mutation was originally selected for in collie-lineage breeds but is rare in true BCs.
- **What it is:** A mutation in the P-glycoprotein gene that affects how the brain blood-brain barrier handles certain drugs. Affected dogs can have severe reactions to ivermectin, loperamide, vincristine, and others.
- **Inheritance:** Autosomal codominant — carriers can show partial sensitivity.
- **Test:** Washington State University VCPL.
- **Action:** Ask the vet to confirm MDR1 status before any ivermectin-class dewormer, anti-cancer drug, or loperamide (Imodium).

3.5 Other less common but known

- **Hip dysplasia.** Multi-genic, polygenic; ~12% incidence in BCs per OFA data. Bucco's nutrition and exercise management through 18 months is the biggest controllable factor.
- **Epilepsy.** Onset typically 6 months-5 years. No single test; familial pattern is the biggest indicator.
- **Hypothyroidism.** Adult-onset; manageable with daily medication.
- **Sensory neuropathy (SN).** Rare; affects working lines slightly more.
- **Goose-stepping / Border Collie collapse (BCC).** Rare; exercise-induced collapse, usually after sustained high-intensity work in adolescent/young adult BCs.

4. The Compulsive Disorder Connection

BCs are the **single most over-represented breed** in canine compulsive disorder (CCD) research, particularly for light-chasing and shadow-chasing.

The science

Karen Overall (Penn Vet) and the Tufts behaviorist group have studied CCD extensively. Findings:

- **Different brain structures.** Affected dogs show measurable differences in glutamate signaling and chromosome 7 markers. Some of this work translated back to human OCD treatment.
- **Heritable.** Repetitive behaviors run in lines. If Bucco's sire or dam shows any compulsive tendency, his risk is elevated.
- **Triggered, then crystallized.** Often starts with an environmental trigger (laser pointer, reflective surface, prey movement) and becomes self-sustaining.

The most common BC compulsions

1. **Light/shadow chasing.** The single most BC-specific compulsion. Begins as casual interest, becomes fixation, becomes uninteruptible. Years of treatment may be needed.
2. **Tail chasing.** Starts cute; ends in self-injury. Often comorbid with anxiety.
3. **Fly snapping / air biting.** Snapping at invisible flies. Can be neurological (seizure variant) or compulsive.
4. **Flank sucking.** More associated with Dobermans but seen in BCs.
5. **Spinning.** Often related to stress or boredom.
6. **Vocal fixations.** Repetitive barking at specific stimuli.

Prevention rules for Bucco – non-negotiable

- **NEVER play laser pointer. NEVER. NOT ONCE.** Even “just for a minute.” The BC brain is the most vulnerable canine brain to this trigger. There are documented cases of dogs developing lifetime fixation from a single session.
- **Interrupt the first instance of staring at reflections, shadows, fan blades, headlight patterns on the ceiling.** Don't wait to see if it escalates.
- **Give him real motor pattern outlets.** Fetch, agility foundations, herding-style play with a Jolly Ball or sheep ball. The brain that gets to run the predatory pattern in legitimate ways is less likely to invent illegitimate ones.
- **Watch for trigger stacking.** Sleep deprivation + boredom + isolation can crystallize a casual interest into a compulsion. Manage all three.

If you see early signs

- Brief staring at reflections → redirect immediately. Don't yell; redirect to a higher-value activity.
- Casual tail-chasing → veterinary behaviorist consult if it persists more than 1-2 weeks.
- Light chasing escalation → veterinary behaviorist NOW. Don't wait. Don't try to handle it with podcasts and Reddit. This is one of the few problems where early intervention is dramatically more effective than late.

5. BC Cognition – Chaser And What She Proved

Chaser the Border Collie (2004-2019), trained by Dr. John Pilley at Wofford College, became the most-studied cognitive subject in canine science.

Documented capacities:

- **Learned the proper names of 1,022 objects.** Each name mapped to a unique toy. Training stopped at 1,022 not because Chaser plateaued, but because Pilley's time ran out.
- **Demonstrated inferential reasoning by exclusion.** Given a familiar set of toys and one new one, then asked for the new toy by name, she retrieved it correctly – inferring that the unfamiliar word must refer to the unfamiliar object.
- **Distinguished verbs from nouns.** “Paw the ball” vs. “Take the ball” – different actions on the same object.
- **Generalized “category” learning.** Could understand that “frisbee” referred to multiple specific objects in the same category.
- **Retained vocabulary for years.** Word knowledge persisted across breaks.

Published in *Behavioural Processes* (2011) and *Learning and Motivation* (2013). Changed the lower bound of what we thought non-primate animals could do with language.

What this means for Bucco

- His capacity is enormous. The ceiling on what he can learn isn't a brain limit – it's a teaching-time limit.
- Words are real cues to him. He's tracking them whether you mean to teach them or not. (“Walk.” “Outside.” “Treat.” Bucco is *already* learning these.)
- Trick training is enrichment. Teaching him 5 named objects, 10 named actions, 3 places, etc., builds vocabulary and engagement – and runs his brain harder than a 2-mile walk.

6. Behavioral Genetics – How Much of Bucco Is Genes?

Scott and Fuller estimated genes ≈ 35% of adult behavior, environment ≈ 65%. Modern behavior-genetics work refines this:

- **Breed-typical motor patterns** (eye-stalk-chase, retrieve, point) – *highly heritable* (70-85%).
- **Reactivity / fearfulness baseline** – moderately heritable (~30-50%).
- **Sociability with humans** – moderate.
- **Trainability** – moderate.
- **Specific learned behaviors** – environmental dominant.

In practice: **Bucco was born with the eye-stalk-chase wiring. He was NOT born with a recall, a sit, or a settle.** The breed-typical traits are nearly fixed; the trained behaviors are almost entirely on you.

This also means: **fearfulness has a heritable component.** A nervous parent dog produces nervous puppies on average, even with great socialization. If Bucco shows nervous tendencies, the response

is the same regardless of cause: positive exposure at distance, classical counterconditioning, time. Don't blame the breeder; manage what's in front of you.

7. The BC Adolescence – Especially Brutal

General canine adolescence runs 6-18 months for medium breeds. BC adolescence is the *peak frustration period* for owners, because:

1. **Trainability dips** (peer-reviewed: Asher et al. 2020). Specifically with the primary handler. The dog WILL “forget” what he knows.
2. **Hormonal arousal climbs**. Testosterone/estrogen spikes raise reactivity and impulsivity.
3. **The predatory motor pattern matures**. What was puppy interest in moving things becomes adolescent fixation. Stalking, staring, chasing all intensify.
4. **Working drive emerges**. The dog wants a *job* and gets frustrated without one. Many BC behavioral problems start here – what looks like “rebellion” is often genuine unmet need.
5. **Second fear period** may hit during this window (typically 8-11 months for medium breeds).
6. **Social politics** matter more. Adult-to-adolescent dog interactions become more complicated. Other dogs may suddenly correct him for behavior they tolerated as a puppy.

Management priorities through adolescence

- **Maintain rather than expand**. Don't introduce new training challenges during the trough.
 - **Channel drive, don't suppress it**. Sport, structured games, herding-style play.
 - **Triple down on classical conditioning** for any developing triggers.
 - **Catch early signs of compulsion**. Adolescence is when fixations crystallize.
 - **Be patient with regression**. It's a stage, not a personality change.
-

8. The Working/Pet Dilemma

A BC bred to work sheep, kept as a pet in an apartment, can thrive – but only with deliberate effort. The breed's drive isn't optional and isn't reduced by long walks. What it needs:

1. **Mental work**. Training sessions, puzzles, scent work, named objects, shaping. 20 min of mental work ≈ 2 hours of walking for arousal management.
2. **Predatory pattern outlet**. Fetch with rules, tug with cues, herding-style ball work, agility.
3. **An off-switch**. Taught deliberately. A BC that's “on” all the time burns out, both behaviorally and physically.
4. **Calm modeling**. Owners who are themselves calm produce calmer BCs. BCs read human arousal with unusual precision.

What to AVOID

- **Repetitive ball-throwing as the primary exercise**. Builds obsessive ball drive without satisfying mental need.

- **Dog parks.** BCs in particular get bullied (small bodies, intense focus draws attention) or develop herding/control issues toward other dogs. Many BC owners report the “first incident” of reactivity traces to a dog park visit.
 - **Long flat-out runs with no thinking.** Tires the body, leaves the brain wired.
 - **Inconsistent routines.** BCs thrive on rhythm; chaos cranks their anxiety.
-

9. The “Velcro Dog” Pattern

BCs bond with intensity. The pattern:

- Selects one primary person (often the one who feeds/trains most).
- Tracks that person around the house, frequently in line-of-sight.
- May develop mild separation anxiety if not deliberately conditioned to be alone.
- Reads micro-expressions of the primary person — frustration registers as anxiety; calm registers as safety.

Implication for Bucco: Scott is likely his primary. Nina is secondary but central. Both need to:

1. **Deliberately train alone time.** Crate or pen with a chew, person leaves the room, returns calmly, no big greeting. Build from 1 minute to 30 minutes over weeks.
 2. **Both train.** A BC who only listens to one person becomes brittle. Nina should run sessions too.
 3. **Regulate own arousal.** When Scott is frustrated, Bucco is frustrated. This is a real thing, biologically — BCs respond to cortisol cues, micro-expressions, and tonal shifts.
-

10. What Working BCs Have That Bucco Won’t (Probably)

Working BCs on a farm get:

- 6-12 hr/day of intermittent stockwork
- Constant predatory motor pattern access in a regulated form
- Open space, environmental variability
- A clear job structure

Bucco won’t get these. That’s fine — millions of pet BCs thrive. But it means you have to *manufacture* the equivalent. The training plan, the sport, the daily enrichment, the puzzles — these aren’t luxury. They’re the apartment-BC equivalent of the working farm.

11. Fun BC Trivia (because Scott asked for fun facts)

- **Chaser knew 1,022 nouns.** Replication studies on other BCs (Rico the German Shepherd, Bunny the Sheepadoodle via AAC buttons) suggest this isn’t unique — the *willingness* to learn is the rate-limiting factor, not capacity.
- **The “BC stare” — known to sheep handlers as “giving eye” — is the dog’s natural method of moving stock without barking.** Other herding breeds (heelers, corgis) use vocal pressure; BCs use silent pressure.

- **The breed standard color is “any color or marking” — but most BCs are black-and-white because the working dogs that founded the breed happened to be that color.** Working breeders never selected for color; they selected for ability.
 - **Old Hemp had a son named Sweep who was famously fast and quiet. Sweep is in most modern working pedigrees.**
 - **The first International Sheep Dog Society trial was held in 1873 in Bala, Wales, won by James Thomson’s dog Tweed.** The trial format became the foundation of modern stock-dog competitions.
 - **A 1994 study (Coren) ranked BCs as the most trainable dog breed.** It’s a flawed metric (it measured “obedience to first command”), but the directional finding has held up.
 - **The “eye” is genetically heritable but trainable.** Some BC lines have “strong eye” (intense, slow-stalking) and others have “loose eye” (more upright, more vocal). Different stock-work styles favor different eye types.
 - **The fastest known BC, Striker, can open a non-electric car window in 11.34 seconds (Guinness, 2004).** Yes, by pawing the crank handle. He learned by watching his owner.
 - **BC herding instinct usually emerges around 6-12 months.** Pups under 6 months who don’t show it haven’t “failed” — the wiring hasn’t matured yet.
-

12. The One Truth That Underlies It All

Border Collies don’t need to work because they’re high energy. They need to work because they have an unsatisfied neural reward circuit that lights up specifically during the eye-stalk-chase sequence.

Give Bucco legitimate outlets for that circuit and he’ll be a remarkable dog. Deny him outlets and his brain will invent illegitimate ones — and those are the problem behaviors people complain about.

That’s the whole game.

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Chapter 6 • Fun Facts & Cool Research

A grab-bag of things that are interesting in their own right, that you can share with Nina or anyone who asks about the new dog, and that reframe what an animal is and what Bucco specifically is capable of.

Cognition

Dogs read human pointing better than chimpanzees do. Brian Hare's foundational work at Duke showed dogs follow human gaze and pointing gestures fluently — chimps mostly don't. This is one of the few cognitive tasks where dogs outperform our closest primate relatives, and it's a product of domestication (wolves don't read pointing well either).

Dogs have functional theory of mind, sort of. They distinguish whether a human is looking at them or not before deciding whether to “steal” food. They behave differently in front of an attentive human than an inattentive one.

Dogs experience genuine separation distress, neurologically. fMRI work (Berns et al. at Emory) on awake, unrestrained dogs showed activation in the caudate nucleus (reward area) when the dog's primary human appeared — the same pattern that lights up when humans see their own children.

Chaser the BC learned 1,022 nouns. Not 1,022 commands. Different nouns for 1,022 different objects, each one a unique referent. And she figured out new ones by exclusion — if you ask for a word she didn't know and there's one unfamiliar object in front of her, she'd retrieve that one.

Bunny the Sheepadoodle, via AAC buttons, has produced multi-button sentences that suggest grammatical sequencing. The science on AAC dogs is still being adjudicated, but the methodology (push-button word boards) at minimum shows dogs can chain words intentionally. (Border Collie / Poodle mix.)

Rico the German Shepherd was the first dog studied scientifically for vocabulary. ~200 words. His work led to Pilley taking on Chaser, who went 5× further.

Dogs dream. REM sleep behavior in dogs maps reliably to dreaming. Puppies dream much more (longer REM stretches, more visible twitching/vocalizing) because they have more new experiences to consolidate.

Sleep

Puppies need 18-20 hours of sleep a day. Not “could use” — *need*. Most “hyperactive puppy” complaints trace to insufficient sleep, not insufficient exercise.

Adult dogs sleep ~12-14 hours/day. Distributed across short bouts rather than one long block.

Dogs have two distinct REM/NREM cycles, like humans. But cycle length is shorter (~20 min vs. 90 min in humans), which is why they nap and wake more frequently.

Puppy REM at 12 weeks is ~30% of total sleep. Adult REM is ~10%. The drop happens slowly over the first 6 months and correlates with the consolidation of new behaviors.

Border Collie Specifics

Old Hemp (born 1893, Northumberland) is the foundation of nearly every modern BC pedigree. He was famous for a quiet, intense working style that became the breed standard for stock-handling.

The BC didn't get AKC recognition until 1995 — over a century after Old Hemp — because working breeders resisted, correctly anticipating that show breeding would split the population into working and non-working lines.

"Strong eye" vs. "loose eye" is a real distinction in working BC genetics. Strong-eyed dogs work with intense, low-stalking pressure. Loose-eyed dogs work more upright and may use voice. Both styles work; different stock and handlers prefer different.

A herding BC can move sheep without ever touching them. The "eye" is so threatening to sheep (in prey-prediction terms — it looks like a stalking predator about to chase) that the sheep move under nothing but visual pressure.

BCs were ranked the #1 most trainable dog breed by Coren (1994). The metric ("obedience to first command in <5 reps") is crude, but the result has held up across different methodologies.

Striker the BC opened a non-electric car window in 11.34 seconds (Guinness World Record, 2004). He learned by watching his owner.

Chaser was a working-line BC, not show-line. The research subject pool for "BCs that learn 1000+ words" comes disproportionately from working lines because they have the focus and drive to sustain three years of daily training.

The BC herding instinct usually shows around 6-12 months. Pups that don't show it at 4 months haven't failed — the wiring isn't mature yet.

Sensory and Physical

Dogs see in color, but limited. Two-cone color vision (blue and yellow) — similar to red-green colorblind humans. A red ball on green grass is harder for them to see than a blue one.

Dogs see ~60 frames per second of motion; humans see ~24. Old CRT TVs flickered for dogs the way fluorescent lights flicker for migraine-prone humans. This is part of why BCs in particular are sensitive to old TVs and screens.

Dogs have ~220 million olfactory receptors; humans have ~5 million. The olfactory bulb in a dog brain occupies a proportionally massive area. They aren't living in our world; they're living in a richer chemical world we can barely perceive.

A dog's sense of smell can detect a single drop of blood in an Olympic-sized swimming pool. Forensic and medical detection dogs work at limits that defy conventional analytical chemistry.

Dogs can smell time. Specifically, they can detect the rate of odor decay, which is why a dog "knows" you're coming home around 5pm — your morning scent is at a specific decay state at that time.

Border Collie heart rate at rest: ~70-100 BPM, lower in trained working dogs.

Dogs can develop calluses on their elbows from hard floors. The “elbow callus” is normal and not painful, but it’s a marker that the dog is consistently lying on hard surfaces — consider providing softer bedding.

Behavior

Yawning is contagious between humans and dogs, with dogs more likely to yawn after their primary human yawns than after a stranger. Possible empathy marker; research is ongoing.

Tail wags aren’t just “happy.” Wag direction and amplitude carry information. - More to the right = positive emotional state (left-brain dominance) - More to the left = negative/avoidance state (right-brain dominance) - Low, slow wag = uncertain - Helicopter wag with whole-body wiggle = full positive arousal

The “play bow” is a meta-signal — it tells the other dog (or human) “what I do next is play, not for real.” Dogs maintain this signal throughout play and it’s the single most reliable indicator that they’re enjoying themselves.

Dogs whose tails are docked or naturally bobbed have measurable difficulty communicating with other dogs. Tail-based signals carry significant social information and dogs without functional tails are at a communication disadvantage.

BCs are statistically more likely than most breeds to engage in displacement behaviors (yawning, lip-licking, scratching) when conflicted. Watch for these — they’re early stress indicators before the dog escalates to growling/snapping.

Training Science

Marker training (clicker) was invented for dolphins in the 1960s by trainers including Karen Pryor. Dolphins can’t be physically reached for treat delivery, so trainers used a whistle marker to indicate “what you just did is correct” — followed by a fish. Pryor adapted the method for dogs in the 1980s.

A study comparing positive-reinforcement and aversive-method training found: dogs trained with aversive methods showed elevated cortisol for hours after sessions, more displacement behaviors during training, and worse performance on novel tasks. Same handlers, same dogs, randomly assigned method.

The 50% acceleration figure for marker training comes from comparative studies on complex behavior acquisition (Smith & Davis, IAABC Foundation Journal). For simple behaviors the advantage is smaller.

A “jackpot” reinforcement (unexpectedly large reward) has been shown to *temporarily increase* the probability of the immediately-preceding behavior much more than steady reinforcement. Use sparingly to mark breakthrough moments.

Domestication

Dogs split from wolves between 20,000-40,000 years ago. The exact date is contested; recent ancient-DNA work suggests two domestication events (one in Europe, one in East Asia) that later merged.

The “fox domestication experiment” (Dmitri Belyaev, Russia, started 1959) selectively bred silver foxes for tameness alone. Within 10-20 generations, the foxes showed dog-like traits: floppy ears, curled tails, piebald coats, juvenile faces. This established that “domestication syndrome” is genetically linked – selecting for one trait (tameness) brings the others along.

Dogs and wolves share 99.96% of their DNA. The differences are concentrated in starch digestion genes, social cognition genes, and some neurodevelopmental timing genes.

Dogs are the only animal that reliably forms cross-species social attachments with humans during a single sensitive period of development. Cats can, but with much more variability. Most other species have to be hand-raised from neonatal stage and even then bond weakly.

Health and Longevity

Border Collies have a typical lifespan of 12-15 years. Working lines often live longer than show lines, possibly because of leaner body condition and lower cancer rates in less-inbred populations.

The Purina lean-dog study (followed Labrador pairs over 14 years; one dog of each pair fed normally, the other 25% less) found lean dogs lived **~1.8 years longer on average** than their ad-libitum littermates. This is one of the cleanest pieces of evidence in canine health research.

Hip dysplasia is multifactorial, not just genetic. Genetics sets the susceptibility; environment determines expression. The single biggest controllable factor in growth: avoiding overfeeding and excessive impact (stairs, jumping off furniture) until growth plates close at ~12-18 months.

Spay/neuter timing has measurable effects on cancer risk and orthopedic disease. The 2020 UC Davis study covering 35 breeds found, for medium and large breeds, that pre-pubertal sterilization increased rates of: cranial cruciate rupture, hip dysplasia, lymphoma, hemangiosarcoma, mast cell tumors, and osteosarcoma. The effect was breed-specific and often striking (golden retrievers spayed before 6 months had ~4× the rate of joint disorders).

Bucco-Specific Curiosities

The name “Bucco” is the Italian word for “fop” or “dandy” – also a Pittsburgh Pirates nickname (from “Buccaneers”). Pittsburgh dog, Pirates dog, fitting on multiple levels.

Bucco’s brain is currently ~70-80% of its adult mass. It will reach ~95% of adult size by 6 months but continue myelinating internal connections until ~18-24 months.

Bucco’s primary socialization window closes around June 8, 2026 (week 16). What he experiences positively between today and then will shape him more than any other 4-week window of his life.

Bucco's 16-week vaccine appointment (June 11) is also a training session. The vet building is now a stimulus that will form a classical association — strongly positive or mildly negative depending on the next ~6 visits. Bring treats. Lots of them.

Adult Bucco's predicted weight: 35-50 lb based on parental size and current growth trajectory. Working-line BCs typically run lean at 35-45 lb. Bring your best guess to the vet for accurate medication dosing.

Bucco was raised with cats from day one. This is unusual and valuable. Most BCs raised without cats develop persistent stalking interest. Bucco's social template will include cats as conspecifics in some sense — he's likely to be safer around them than the average BC, *as long as* the leashed-around-cats protocol holds through adolescence.

The Last One

Dogs are the first species humans deliberately co-evolved with. Everything else came later — livestock, cats, horses. The relationship between humans and dogs is the foundational case study for inter-species cooperation. When you sit on the floor and teach Bucco “sit” using kibble and a marker word, you're participating in a 20,000-40,000 year old behavioral exchange.

That's pretty cool.

Sources for the curious

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- [Hare et al. — The domestication of social cognition in dogs \(Science 2002\)](#)
- [Coren — The Intelligence of Dogs \(1994\)](#)
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Chapter 7 • Warnings & Red Flags

A focused document on what NOT to do, what to watch for, and when to get professional help. Some of this is breed-specific. Some of it applies to any puppy.

1. The Never List (Border Collie-specific)

These are things that, once done, are hard or impossible to undo.

1.1 Laser pointers

Never. Not once. Not “just for a minute.”

BCs are the single most-susceptible breed to compulsive light fixation. There are documented cases of dogs developing lifetime light-chasing obsession from one play session. The fixation looks like fun at the time. Then it persists when the laser is off — staring at reflections, fan shadows, headlight patterns on the ceiling, light fluctuations from leaves. Years of veterinary behavioral intervention may be required.

There is no safe BC use case for laser pointers. There is no version of “I’ll be careful.” If a guest pulls one out, you take it.

1.2 Reflective surface play

Mirrors, polished metal, water reflections, screen reflections. If Bucco shows interest, redirect immediately to a higher-value activity. Don’t laugh, don’t engage. The first time is curiosity; the fifth time is a habit; the fiftieth is a pathology.

1.3 Dog parks

A dog park is an unstructured space with random dogs of varying social skills, varying training, and varying owner attentiveness. BCs are particularly poorly served by this environment for three reasons:

1. **They get bullied.** Smaller frame, intense focus, often reads as “weird” to dogs of less sensitive breeds. Easy target.
2. **They invent herding/control behaviors.** The BC who can’t stop other dogs from running becomes the BC who tries to control them with nips and body-blocks. This generalizes to all dogs.
3. **They learn that other dogs are scary or chase-worthy.** Either outcome is bad and both are common.

Structured puppy class > dog park. Always. Forever. Bucco can be 4 years old and well-socialized and a dog park is still a bad idea for him.

1.4 Repetitive ball throwing as primary exercise

A BC will retrieve until physical collapse. The brain gets dopamine pulses from the chase phase, but never from a settle/end signal — so the dog spirals into hyper-arousal rather than tiring. You can

produce a chronically anxious, obsessive ball-driven BC by this single training pattern.

Use fetch with structure: 5-10 retrieves, settle on cue, drink water, end on calm. Not 45 minutes of nonstop throwing.

1.5 Sustained running with no thinking

Marathon walks, treadmill running, bike-jogs without breaks. Tires the body, leaves the brain wired. Worse: it builds physical conditioning that then *demands* more activity for “tiredness.” You’ll be hiking 4 hours a day at year 3.

Instead: 20-min training session → calm settle → 30-min decompression walk → calm settle → puzzle feeder. The puppy is *more* tired with less physical exertion when the brain has been engaged.

2. The Never List (Universal Puppy)

2.1 Adult dog food during growth

Adult food doesn’t guarantee the growth-phase calcium/phosphorus profile. Feeding it during growth can cause skeletal disorders.

2.2 Calcium, vitamin D, or vitamin A supplementation

Beyond what’s in the food. All three cause harm in excess.

2.3 Cooked bones (any kind)

They splinter. Splinters perforate the esophagus, stomach, intestine. Raw meaty bones under supervision, after adult teeth are fully in (~6-7 months), is a different conversation — but never cooked.

2.4 Sudden food changes

Always 7-10 day transitions. The microbiome needs time.

2.5 The toxic-food list

Food	Why
Grapes & raisins	Acute kidney failure, mechanism unknown, no safe dose
Onions, garlic, chives, leeks	Hemolytic anemia
Chocolate (especially dark)	Theobromine + caffeine toxicity
Xylitol	Insulin release → severe hypoglycemia (deadly)
Macadamia nuts	Neurological symptoms
Avocado	Persin (mostly an issue in large quantities)
Yeast dough	Ferments in stomach
Alcohol	Liver, kidney, neurological
Cooked bones	Perforation
Apple cores, peach pits, cherry pits	Cyanide compounds

Food	Why
Coffee, tea, energy drinks	Caffeine toxicity
Salt (large amounts)	Sodium ion poisoning
Pizza dough, raw bread dough	Yeast + alcohol fermentation
Marijuana	THC toxicity (and worsening with edibles)

2.6 Stairs and furniture jumping under 6 months

Growth plates are open. A bad landing this month → orthopedic problem at year 3 or 4. Carry the puppy on stairs. Block jumps off the couch.

2.7 Forced exercise during growth

The 5-minute rule: 5 minutes of structured leash walking per month of age, up to twice a day. So an 8-month-old gets max 40-minute structured walks. Free play in safe spaces is different — the puppy self-regulates.

This rule is conservative; some studies don't show clean cause-effect at moderate exercise levels. But for a BC with a working drive that will *not* self-regulate, the rule is protective.

2.8 “Alpha rolling” or any dominance-based correction

Decades of research have eroded the dominance model of dog training. The “alpha wolf” framework Cesar Millan popularized was based on captive-wolf studies that were later debunked by the researcher who did them (David Mech, who literally wrote the book and has spent decades trying to retract his earlier work). Forcing a dog onto its back, pinning, or “showing dominance” is at best ineffective and at worst dangerous (you'll get bit).

2.9 Leash corrections (pops/jerks)

They suppress behavior in the immediate context without teaching what to do instead. They damage the human-dog relationship. They produce cortisol responses. There is no version of this that's appropriate for Bucco's training plan.

2.10 E-collars / shock collars / prong collars

Same logic as leash corrections plus equipment-specific damage. The training plan doesn't need them. Anyone telling you it does is selling you something or hasn't kept up with the science.

3. Behavioral Red Flags — Get Help Now, Not Later

These are behaviors that respond *dramatically* to early intervention (days to weeks) and very poorly to late intervention (months to years). Don't try to handle these alone.

3.1 Resource guarding

Early signs (immediate intervention): - Stiffening over a food bowl, toy, chew - Lip lift / showing teeth when approached over a valued item - Quick gulping of food when someone walks near - Possessive of a spot (couch, bed) with low growl when approached

Why intervene fast: resource guarding intensifies with every successful occurrence. By the time it's a bite, you're months into reconditioning.

First aid: trade-up protocol. Never just take things away. Always offer a higher-value swap. Walk by the food bowl during meals and drop in something amazing. Make humans approaching valued items a *prediction of more value*, not less.

Get help: if growl, bared teeth, snap, or bite has happened, call a KPA-CTP or CPDT-KA certified trainer — or a board-certified veterinary behaviorist if it's already at snap/bite stage.

3.2 Reactivity

Early signs: - Barking and lunging at other dogs on leash - Stiff posture, raised hackles, intense stare at triggers - Difficulty disengaging from a stimulus - "Trigger stacking" — calmer stimuli setting him off after a hard day

Why intervene fast: reactivity is self-reinforcing. The dog barks → the trigger goes away (because the dog walking past keeps walking) → the dog learned barking works.

First aid: distance. Never put him over threshold. If he's reacting, you're already too close. Cross the street, U-turn, create distance. Feed at sub-threshold distance to build a positive emotional response.

Get help: if it's persistent or escalating beyond 2-3 weeks of management.

3.3 Compulsive behaviors

Early signs: - Staring at lights, shadows, reflections - Tail chasing that doesn't stop on a redirect - Snapping at invisible flies / air biting - Flank sucking / self-licking that creates skin lesions - Repetitive pacing in patterns

Why intervene fast: these can become neurological over months. The treatable window is short.

First aid: interrupt immediately and redirect to a high-engagement activity. Document frequency and context.

Get help: veterinary behaviorist NOW for any persistent compulsion. Penn Vet, NC State, Cornell, Purdue, and UC Davis all have behavior services that accept referrals.

3.4 Severe noise phobia

Early signs: - Trembling, hiding, drooling at thunder/fireworks - Refusing to eat for hours after a noise event - Generalizing fear to weather changes (barometric pressure, wind)

Why intervene fast: sound phobias generalize and worsen with each exposure. The dog who fears thunder this year fears wind next year and stairs the year after.

First aid: sound desensitization with food during meals at LOW volume. Pair scary sounds with extraordinary reinforcement.

Get help: anti-anxiety medications (Sileo, trazodone, gabapentin) are available and well-tolerated; ask the vet. Behavior modification + medication together produces much better outcomes than either alone.

3.5 Bite incidents (not nipping)

A bite is a tooth-on-skin event with intent or force. Nipping during play with no bruise or break is normal puppy behavior. A bite that breaks skin, leaves bruising, or is delivered with intent (not play) is different.

Get help immediately. This is not a “wait and see.” Document: who, what, when, where, what was the dog doing 60 seconds before, what was the human doing 60 seconds before.

3.6 Separation distress that doesn't resolve

Some puppy fussing on isolation is normal. *Persistent* distress that doesn't resolve with proper crate conditioning over 2-3 weeks is a different problem.

Signs: - Salivating, panting, drooling when left alone - Destruction focused on exit points (doors, windows) - House-soiling only when alone - Refusing to eat when alone

Get help: before it worsens. Separation anxiety has good treatment protocols, but they require expertise.

3.7 Sudden behavior change

A dog that was fine and now isn't. Sudden aggression, sudden fear, sudden lethargy, sudden change in appetite. **The first stop is the vet, not the trainer.** Pain, hormonal changes, neurological issues, and metabolic problems can all present as “behavior change.”

4. Medical Red Flags — When To Call The Vet

4.1 Emergency (go now)

- Bloated, hard abdomen + non-productive retching (GDV / bloat — life-threatening, deep-chested breeds at higher risk but ANY dog can develop it)
- Suspected toxin ingestion (any of the toxic foods, medications, chemicals, certain plants)
- Heatstroke signs: heavy panting, drooling, weakness, collapse, body temp >104°F
- Hives, facial swelling, vomiting after a vaccine or insect sting (anaphylaxis)
- Sustained seizures or seizures lasting >2 minutes
- Significant trauma — hit by car, dropped, attacked
- Heavy or persistent bleeding
- Inability to urinate or defecate

4.2 Urgent (today)

- Vomiting more than 2-3× in a few hours
- Diarrhea with blood
- Refusing food for >24 hr (puppy under 6 months: <12 hr is concerning due to hypoglycemia risk)
- Significant lethargy out of character
- Limping that doesn't resolve in a few hours
- Painful abdomen on gentle touch
- Excessive water drinking (polyuria/polydipsia)

4.3 Soon (within a few days)

- Loose stool persisting >48 hr
- Coughing, sneezing more than occasional
- Skin issues (itching, hot spots, lesions)
- Ear shaking, head tilt, scratching at ears
- Eye discharge or squinting

4.4 At Bucco's age — extra-vigilant on

- **Parvo.** Even fully vaccinated puppies (until ~2 weeks post-final shot, ~late June 2026) are at residual risk. Signs: bloody diarrhea, vomiting, lethargy, refusal to eat. Mortality without treatment is very high. ER immediately if suspected.
 - **Hypoglycemia.** Under 6 months, prolonged fasting can drop blood sugar dangerously fast. Honey or Karo syrup on the gums + immediate vet visit if you find him weak, wobbly, or unresponsive.
 - **GI obstruction.** Puppies eat things. Socks, hair ties, rocks, toys. Vomiting + lethargy + abdominal pain = vet now, not tomorrow.
-

5. Vaccine Reactions

5.1 Normal post-vaccine (next 24-48 hrs)

- Mild lethargy / extra sleepiness
- Slight swelling at injection site
- Low-grade fever (you won't measure but you'll see a sleepier dog)
- Reduced appetite for one meal
- Slight soreness at injection site

5.2 Concerning (call the vet)

- Persistent vomiting
- Diarrhea
- Soft swelling at injection site persisting >5 days

5.3 Emergency (ER now)

- **Hives** — bumps under the coat anywhere on body
- **Facial swelling** — especially around eyes/muzzle
- **Vomiting + collapse / weakness**
- **Difficulty breathing**
- **Severe lethargy + pale gums**

For Bucco: the June 11 vaccine visit is a candidate for these. Watch the first 4 hours closely.

6. The Things That Look Bad But Usually Aren't

(So you don't panic at the wrong things.)

- **Hiccups.** Normal in puppies; they grow out of it.
 - **"Frenetic Random Activity Period" (zoomies).** The 8pm puppy explosion is normal energy release. If it always happens at the same time of day and resolves in 5-10 minutes, it's fine.
 - **Stinky breath in a teething puppy.** Adult teeth coming in disrupt the gum line. Some smell is normal. Persistent rotting-fish smell or visible gum disease is not.
 - **One or two soft poops in a busy week.** Diet variation, mild stress, or just normal variability. Persistent loose stool >48 hr is different.
 - **Reverse sneezing** (a honking, snorting inhalation). Looks alarming. Almost always benign. Gently cover one nostril for a few seconds to break the spasm.
 - **Slight injection site swelling.** Normal up to several days.
 - **Mild "puppy regression"** post-vaccine (3-5 days of clumsier training). Normal cortisol response to immune activation.
 - **A single skipped meal** in a 12-week-old. Watch the next meal. If two in a row, call the vet.
-

7. The "Right Person For Help" Decision Tree

Problem →

Medical? → VETERINARIAN

Behavioral but mild and recent? → KPA-CTP or CPDT-KA certified trainer using +R methods

Behavioral with bite history? → VETERINARY BEHAVIORIST (DACVB)

Behavioral + suspected medical component? → VET FIRST, then DACVB or trainer

Sport-specific (agility, herding)? → Sport-specific instructor with credentials in the sport

Compulsive behavior? → DACVB (don't pass go, don't try anything else first)

Pittsburgh-area resources to identify *before* you need them

- **Penn Vet Behavior Service** (Philadelphia) — accepts referrals; long waitlists. Get on it before the crisis if you suspect something brewing.
- **NC State Veterinary Behavior** — Raleigh; another option for referral.
- **Local KPA-CTP / CPDT-KA trainers** — search the CCPDT directory for certified positive-reinforcement trainers near North Hills.
- **Susan Garrett's DogsThat community** — global; can recommend trainers in your area.

- **AVSAB (American Veterinary Society of Animal Behavior)** — position statements and trainer-finder.
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8. The Single Most Important Sentence

Early intervention works much better than late intervention for behavioral problems. Trust your gut. If something seems off, get an opinion before it becomes a pattern.

The cost of an early trainer consult: \$100-200 and an evening.

The cost of a late veterinary behaviorist consult after a serious bite incident: \$400-1000 + 6-12 months of remediation + a relationship and home that may never feel the same.

Cheap on the front end. Expensive on the back end. The math is unambiguous.

9. The Things The Internet Will Tell You That Are Wrong

- “BCs need 4 hours of exercise a day or they’ll go crazy.” Mental work beats physical work. 20 min of training tires a BC more than 2 hours of running. Both have value, but the training is the more sustainable answer.
 - “He’ll grow out of it.” Most behavioral issues *intensify* through adolescence, not improve. If you’d consider it a problem at 3 months, it’ll be a bigger problem at 18 months. Address it now.
 - “Just dominate him; show him who’s boss.” See §2.8. The dominance model is debunked. The handlers who insist on it loudest are usually the ones with the worst-behaved dogs.
 - “Stop using treats; he should obey because he respects you.” Behavioral science is unambiguous: behavior persists when reinforced. Withhold reinforcement and behavior fades. Treats are the most efficient and humane reinforcer for most situations. They aren’t a crutch; they’re the technology.
 - “Mix some adult food in early to teach his digestion.” No. Adult food doesn’t have the growth-phase nutrient profile. Stick to a good puppy or all-life-stages-with-growth food.
 - “Walk him for an hour twice a day or he’ll be hyperactive.” Puppy under 6 months: don’t do this. See §2.7.
 - “Let other dogs correct him; he needs to learn.” Letting an unknown dog “correct” your puppy is a great way to give him a lifetime fear of dogs. The “correction” might be appropriate or might be an over-the-top aggressive response from a poorly-socialized adult. You don’t get to choose; you only see the outcome.
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10. The Last Warning

The biggest risk to Bucco isn’t disease, isn’t injury, isn’t bad genetics. It’s accumulated tiny errors that compound over 18 months of adolescence.

A leash pop here. A laser pointer “just for fun” there. A skipped socialization week. A few extra calories that build into a 15% overweight 1-year-old. A reactivity event left unaddressed. None of these is catastrophic alone. All of them compound.

The compounding works in the other direction too. A handful of right choices every day, repeated daily, builds a dog who is fundamentally different at 18 months from one whose handler made a handful of wrong choices.

Compound interest applies to puppy training.

That’s it.

Sources for the cautious

- [AVSAB Position Statements \(positive reinforcement\)](#)
- [AAHA Behavior Management Guidelines](#)
- [Hart et al. — Long-Term Health Effects of Neutering Dogs \(UC Davis 2020\)](#)
- [FDA — Toxic foods for dogs](#)
- [ASPCA Animal Poison Control](#)
- [DACVB — American College of Veterinary Behaviorists directory](#)
- [CCPDT — Certified Professional Dog Trainer directory](#)